

FINITE-DEPTH ERROR CERTIFICATES FOR RECURRENT EVALUATORS IN TINY RECURSIVE MODELS

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ABSTRACT

We study the fixed-snapshot error of a finite-depth recurrent value evaluator inside a discounted plan-edit MDP. For any two finite depths $m > n$, the depth- n error is bounded by its discrepancy from the depth- m evaluator plus the depth- m Bellman residual divided by $1 - \gamma^K$. A Lipschitz value head controls the discrepancy by the recurrent path length from n to m . This result permits transient computation and assumes no latent fixed point; a contractive fixed-point bound follows only as a specialization. Persistent latents have a separate direct certificate on the augmented state (x, y, z) that also needs no slow drift. Exact statewise centering and direct deployment of the exact policy mixture are required to propagate these errors through conservative policy improvement. A 64-state finite-MDP calculation is a numerical unit test of the composed inequalities. Our empirical scope is limited to 4×4 Sudoku. A persistent implementation obtains 57.4% success over 10 seeds versus 32.0% for TRM+PPO at the same nominal outer-step count but unequal environment interactions; this is neither theorem validation nor evidence of sample efficiency. The theory-aligned episodic configuration obtains 0% over 10 seeds. In a controlled factorial, projection has a +13.0 percentage-point algorithmic effect, while contraction’s +2.2-point effect is inconclusive. These results separate a finite-depth certificate from the current implementation evidence and its limitations.

1 INTRODUCTION

Tiny Recursive Models (TRMs) (Jolicoeur-Martineau, 2025) obtain predictions through iterative latent refinement. In checker-guided tasks such as Sudoku, the same recurrent computation can serve as an internal evaluator for candidate edits to a partial plan. A finite unroll can be useful while remaining transient: there is no general reason for an arbitrary TRM to be designed around convergence to a latent fixed point. Standard approximate-policy-iteration and conservative-policy-improvement (CPI) analyses (Bertsekas & Tsitsiklis, 1996; Kakade & Langford, 2002) place evaluator error in a black-box term and therefore do not distinguish a finite recurrent path from a Bellman residual at a chosen reference depth.

We analyze Unrolled Policy Iteration with TRMs (UPI-TRM), which retains the inner recurrence, uses a scalar value head to score candidate plan edits, and places environment transitions only in the outer checker-guided MDP. The analysis fixes the MDP, recurrence, value head, initialization, and policy pair. It is not an end-to-end theorem for SGD, TD learning, BPTT, or distillation. Euclidean projection can supply forward invariance, and exact baseline summation can supply the statewise centering needed by the CPI step, but these are conditions on a frozen snapshot rather than guarantees about how training reaches it.

Our primary result (Theorem 6.1) compares $U_n(s) = V_\psi(z^{(n)}(s), x)$ with any deeper finite reference U_m for $m > n$. It bounds $\|U_n - V^\pi\|$ by $\|U_n - U_m\|$ plus the Bellman residual of U_m divided by $1 - \gamma^K$, and controls the first term by the recurrent path length for a Lipschitz head. No recurrent convergence is assumed. A forward-invariant contraction geometrically bounds that path and recovers the fixed-point formula as $m \rightarrow \infty$. Corollary 6.3 transfers the value bound to centered advantage error, and Theorem 6.5 supplies a one-step CPI statement only for exact centering and direct deployment of the exact mixture.

The evidence has two distinct roles. First, an exactly solvable finite MDP numerically unit-tests the composed value, advantage, and CPI inequalities; it is not an empirical discovery. Second,

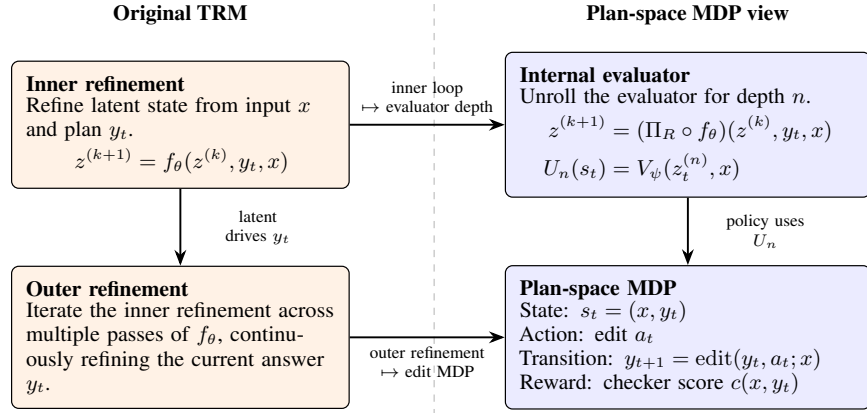


Figure 1: TRM-to-MDP bridge used throughout the paper. Left: the original TRM’s inner latent-refinement loop nested inside its outer continuous refinement of the current answer y_t (Jolicoeur-Martineau, 2025). Right: we swap TRM’s continuous outer refinement for a discrete plan-edit loop under checker feedback, modeling it as a plan-space MDP over instance–plan states, and treat the inner loop as an internal evaluator that is unrolled for depth n . The checker maps to reward; the inner loop maps to evaluator computation inside a single state evaluation.

experiments on hard 4×4 Sudoku evaluate implementations. The theory-aligned episodic run obtains 0% success over 10 seeds. A persistent, distilled implementation obtains $57.4\% \pm 12.2\%$, versus $32.0\% \pm 15.3\%$ for TRM+PPO at the same nominal outer-step count, but their environment-interaction counts differ. We therefore claim neither equal-interaction superiority nor sample efficiency. Projection’s $+13.0$ -point factorial effect combines forward-pass clipping with changed training dynamics; contraction’s $+2.2$ -point effect is inconclusive.

Scope. Episodic latents are Markov on (x, y) . At a fixed snapshot, persistent latents are Markov on (x, y, z) when every transition-relevant variable is included. Their direct residual certificate needs no latent fixed point or slow drift; slow drift is needed only for the stronger comparison with a memoryless fixed-point reference. For the headline run, the uniform augmented residual and exact augmented-state centering are not established, several transition details cannot be audited from the retained package, and distillation is known to violate exact-mixture deployment. The current empirical domain is only 4×4 Sudoku.

Contributions.

- A fixed-point-free finite-reference-depth value and path decomposition, with the contractive fixed-point result as a specialization (Theorem 6.1).
- A direct finite-depth certificate for persistent state on the augmented MDP, separated from an optional slow-drift comparison (Corollary B.1).
- A conditional bridge from these value errors to CPI under exact centering and exact-mixture deployment (Theorem 6.5); this is a fixed-snapshot result, not a training theorem.
- A numerical finite-MDP unit test and a 4×4 Sudoku study that reports the theory-aligned 0% result, the persistent implementation result under unequal interactions, and controlled projection–contraction effects with their limitations.

2 RELATED WORK

Recursive latent reasoning models. TRMs (Jolicoeur-Martineau, 2025) and Hierarchical Reasoning Models (Wang et al., 2025a) use iterative latent refinement for structured reasoning; follow-up work studies training dynamics and test-time adaptation (Qasim & Zhang, 2025; McGovern, 2025). We keep TRM’s inner latent refinement but swap its supervised outer loop for a discrete plan-edit loop under checker feedback, and analyze how finite inner evaluation affects policy improvement. The

contraction assumption is DEQ-adjacent (Bai et al., 2019) only in providing a regime with bounded truncation bias.

Policy-improvement analyses. CPI, TRPO, and safe/monotone policy improvement (Kakade & Langford, 2002; Schulman et al., 2015; Pirota et al., 2013; Metelli et al., 2021; Eshwar et al., 2025) all bound improvement under an abstract evaluation-error term, but treat that error as a single black-box quantity. The closest formal neighbor is function-approximation CPI (Metelli et al., 2021; Eshwar et al., 2025), which relates a learned V_ψ to V^π via a regression model, a sampling distribution, and an optimizer. We do something orthogonal: conditional on sup-norm residual and centering control, we expose the finite recurrent path between depths n and m . A contractive recurrence turns this path into an L_z^n specialization, but contraction is not required by the primary result. Empirical RLVR work on LLM reasoning (Wang et al., 2025b; Su et al., 2025; DeepSeek-AI, 2025) uses verifiable rewards in a similar spirit but does not analyze approximation error. A quantitative join with FA-CPI on V_ψ would need coverage and uniform-convergence assumptions we leave out.

Planning and structured-decision neighbors. Sequence-model RL ranges from architecture-level planning (Tamar et al., 2016) and inference-time search (Janner et al., 2021) to pure sequence modeling (Chen et al., 2021; Laskin et al., 2023); structured prediction and Sudoku/constraint RL (Daumé III et al., 2009; Ross et al., 2011; Chang et al., 2015; Mehta, 2021; Klein, 2022) are closer to our plan-space MDP framing. The TRM inner unroll differs from both: it is architecture-internal computation within one state evaluation.

3 BACKGROUND AND SETUP

The original TRM, given input x and plan y_t , runs an inner loop that updates a latent $z^{(k+1)} = f_\theta(z^{(k)}, y_t, x)$. After n steps it reads out a refined prediction from a task head trained on checker-labeled puzzles (Jolicoeur-Martineau, 2025). UPI-TRM keeps f_θ but replaces the task head with two pieces. The first is an optional projection Π_R that keeps latents inside a radius- R ball (Assumption 5.1, Eq. 23), so the projected update is $z^{(k+1)} = (\Pi_R \circ f_\theta)(z^{(k)}, y_t, x)$. The second is a scalar value head V_ψ that produces the unrolled evaluator output $U_n(s_t) = V_\psi(z_t^{(n)}, x)$ with $s_t = (x, y_t)$, which scores candidate plans in the outer CPI loop.

In the plan-space MDP the external state is $s_t = (x, y_t)$, the action is an edit a_t , and the transition is $y_{t+1} = \text{edit}(y_t, a_t; x)$. A checker score $c(x, y)$ supplies reward, optionally via potential shaping (Ng et al., 1999) $r = r_0 + \gamma\Phi(s') - \Phi(s)$ with $\Phi = c$. The latent z is *not* part of the environment state in our primary analysis; it is internal computation used to evaluate s_t . The resulting process is a discounted MDP $(\mathcal{S}, \mathcal{A}, P, r, \rho, \gamma)$ with $\gamma \in (0, 1)$ and $K \geq 1$. All theoretical statements fix one parameter snapshot: the MDP, recurrence, initialization, value head, current policy π , and candidate policy π_{cand} do not change while the displayed quantities are evaluated. Rewards are bounded, policies and evaluator outputs are measurable, and the relevant value functions are bounded on a measurable advantage-evaluation closure that is invariant under π almost surely. For finite Sudoku these conditions hold automatically; otherwise all suprema are essential suprema on the stated closure sets. Increasing n spends more computation inside one state evaluation but does not change the environment horizon or search depth, so the analysis tracks evaluator truncation error inside a fixed external MDP.

Episodic- z vs. persistent- z . In *episodic- z* , the latent is reinitialized from the current plan at each edit step, so (x, y) remains a Markov state. In *persistent- z* , the latent is carried across edits, matching the original TRM. With persistent latents the next-state distribution depends jointly on (x, y, z) , since the same plan paired with a different carried latent can produce a different next plan after another inner unroll. Treating (x, y) as the state in this regime is a non-Markov approximation. Appendix B lifts the finite-depth residual certificate directly to $\bar{s} = (x, y, z)$; this step needs no latent fixed point or slow drift. A separate slow-drift assumption is required only for comparison with a memoryless fixed-point reference.

Notation. For one common initialization $z^{(0)}(s) = z_{\text{init}}(s)$, let $z^{(j)}(s)$ be the latent after j applications of the frozen recurrent map and let $U_j(s) = V_\psi(z^{(j)}(s), x)$. At the absorbing state, every U_j uses the same depth-independent normalization from Appendix J. Only the optional contractive specialization uses the latent fixed point z^* and $U_*(s) = V_\psi(z^*(s), x)$. We write

Algorithm 1 One UPI-TRM update on the plan-space MDP

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1: Initialize: sample instance  $x$  and initialize plan  $y_0$ 
2: Latent source: reinitialize from the current plan (episodic- $z$ ) or carry from the previous edit (persistent- $z$ )
3: for  $t = 0, \dots, T - 1$  until solved or budget exhausted do
4:   Evaluator:  $z_t^{(n)} \leftarrow (\Pi_R \circ f_\theta)^{\circ n}(z_t, y_t, x)$ 
5:   Action:  $U_n(s_t) \leftarrow V_\psi(z_t^{(n)}, x)$ ; sample  $a_t \sim \pi_\phi(\cdot \mid y_t, z_t^{(n)}, x)$ 
6:   Transition: apply the edit, observe  $r_t$ , and store  $(s_t, a_t, r_t, s_{t+1})$ 
7: end for
8: Value update: form the  $K$ -step bootstrap target  $G^{(K)}$  and regress  $U_n$  toward it
9: Advantage estimate: compute centered  $\hat{A}$  by exact baseline summation
10: Policy update: take a policy-gradient step to obtain  $\pi_{\text{cand}}$ 
11: Conservative step:  $\pi_{\text{new}} \leftarrow (1 - \alpha)\pi_\phi + \alpha\pi_{\text{cand}}$ 
12: Optional distillation: distill  $\pi_{\text{new}}$  into a single policy network

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$\|f\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} := \text{ess sup}_{s \in \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} \cup \{s_{\text{abs}}\}} |f(s)|$ for the sup-norm over the measurable advantage-evaluation closure used by CPI. This closure contains states reachable under π and the one-step successors needed to evaluate actions supported by π and π_{cand} , and is invariant under subsequent π dynamics almost surely (Appendix I; full notation in Appendix H).

4 UPI-TRM ALGORITHM

Algorithm 1 alternates three phases: collect edit trajectories, fit U_n against a K -step bootstrap target, and take a conservative mixture step. The formal guarantee applies to the explicit mixture; distillation is an implementation step outside the theorem.

The direct episodic and augmented-state applications of Theorem 6.5 require exact statewise baseline summation, the explicit mixture step, and no distillation. In the persistent case, centering and the mixture identity are conditional on the full augmented state. The explicit mixture (rather than a clipped surrogate, trust-region update, parameter interpolation, or distilled approximation) is required for the linear-in- α tightening; otherwise the centering defect and deployment mismatch lie outside the stated bound.

Persistent- z is handled in Appendix B, and the initialization condition in Appendix N gives $C_z \leq 2R$. Per update, the dominant cost is $O(T \cdot |\mathcal{A}| \cdot n)$ inner-evaluator unrolls: T states per trajectory, $|\mathcal{A}|$ unrolls each for exact baseline summation, and depth n per unroll.

5 BELLMAN RESIDUALS AND AN OPTIONAL CONTRACTIVE REGIME

We train U_n by K -step bootstrapped regression with loss $\mathcal{L}_{\text{val}} = \mathbb{E}_{s_0 \sim \mu} [(V_\psi(z^{(n)}(s_0), x_0) - \text{sg}(G^{(K)}))^2]$, where $G^{(K)} = \sum_{k < K} \gamma^k r_k + \gamma^K \bar{V}(s_K)$ is the K -step bootstrap target. The K -step Bellman operator $\mathcal{T}_K^\pi$ is the standard γ^K -contraction with fixed point V^π (Sutton & Barto, 2018; Puterman, 1994).

The primary finite-reference-depth result in Section 6 uses Bellman contraction but does not assume that the recurrent computation converges. It assumes sup-norm residual control on the relevant closure; we do not prove a general $L^2(\mu) \rightarrow \|\cdot\|_\infty$ conversion. A separate, optional assumption below makes a latent fixed-point specialization available. Bellman contraction is operator-level; inner-evaluator contraction of $z \mapsto \tilde{f}_\theta(z, y, x)$ controls a recurrent path within one frozen forward pass and by itself gives no TD- or training-convergence guarantee.

Assumption 5.1 (Forward-invariant contraction). The latent space $\mathcal{Z}$ is a complete normed space (in our implementation, $\mathbb{R}^d$ with the Euclidean norm). There exists a closed set $\mathcal{Z}_{\text{inv}} \subseteq \mathcal{Z}$ such that the projected update map $\tilde{f}_\theta(z, y, x) := (\Pi_R \circ f_\theta)(z, y, x)$ maps $\mathcal{Z}_{\text{inv}}$ to itself and is an L_z -contraction in z with $L_z \in (0, 1)$. For every (x, y) , the initialization satisfies $z_{\text{init}}(x, y) \in \mathcal{Z}_{\text{inv}}$. Let $z^*(x, y)$ denote the unique fixed point and assume

$$C_z := \sup_{(x, y)} \|\tilde{f}_\theta(z_{\text{init}}(x, y), y, x) - z_{\text{init}}(x, y)\| < \infty.$$

We enforce Assumption 5.1 via operator-norm clamping on $z \rightarrow z$ layers plus per-layer scaling; Π_R provides forward invariance. Value-head spectral normalization is disabled in contraction sweeps to avoid confounds (Appendix D.2). The assumption identifies a regime with rigorous bounds; it is sufficient, not necessary. Appendix D.7 shows that the spectral-norm target does not reliably control achieved post-projection contraction once projection is active, so its practical effect must be measured empirically.

In the projection-active sweep cell of Appendix D.7, $\hat{L}_z^{\text{post}}$ is set mostly by the saturation factor R/ρ_R and stays in a narrow 0.219–0.242 band (Table 11), so the L_z -dependent terms behave as proxies rather than as a tunable knob. A pre-registered episodic- z hard-suite rerun (Appendix G.3) yields a broader post-projection distribution including seeds with $\hat{L}_z^{\text{post}} \geq 1$, so the narrow-band behavior is not a general hard-suite geometry fact.

6 FINITE-DEPTH EVALUATION ERROR DECOMPOSITION AND CPI PENALTY

We connect finite evaluator depth to CPI. The CPI shell is standard (Kakade & Langford, 2002); the TRM-specific content is a finite-reference-depth error decomposition that does not require a recurrent fixed point. We focus on episodic- z and defer persistent- z to Appendix B.

6.1 VALUE ERROR BOUNDS (EPISODIC- z)

K -step Bellman contraction gives, for any bounded measurable V on the forward-invariant closure,

$$\|V - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{1}{1 - \gamma^K} \|V - \mathcal{T}_K^\pi V\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}, \quad (1)$$

and therefore permits any finite unroll depth to serve as the reference.

Theorem 6.1 (Finite-reference-depth value decomposition). *Fix one parameter snapshot, policies (π, π_{cand}) , recurrent map, initialization convention, $K \geq 1$, and a latent norm on $\mathcal{Z} \subseteq \mathbb{R}^d$. Let $V_\psi : \mathcal{Z} \times \mathcal{X} \rightarrow \mathbb{R}$ be the scalar value head. For any finite integers $0 \leq n < m$, let $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} \cup \{s_{\text{abs}}\}$ be measurable, contain the one-step successors required by both policies, and be invariant under subsequent π transitions almost surely. Assume rewards are bounded there; every latent iterate $z^{(j)}$ is measurable; $U_j(s) = V_\psi(z^{(j)}(s), x)$ is bounded and measurable there for $j \in \{n, \dots, m\}$; and $U_j(s_{\text{abs}}) = -C_{\text{max}}$ for every depth. Then*

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \underbrace{\|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}}_{\text{finite-depth discrepancy}} + \underbrace{\frac{\|U_m - \mathcal{T}_K^\pi U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}}{1 - \gamma^K}}_{\text{reference-depth residual}}. \quad (2)$$

If, in addition, the scalar head is L_V -Lipschitz in its latent argument uniformly in x , then

$$\|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq L_V \sum_{j=n}^{m-1} \text{ess sup}_{s \in \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}} \|z^{(j+1)}(s) - z^{(j)}(s)\|. \quad (3)$$

The theorem is a residual bound plus a telescoping path bound; we make no novelty claim for the triangle inequality itself. Its role is to expose a certifiable finite reference m : the recurrence may perform useful transient computation and need not converge before, at, or after m . The same frozen policy π appears in V^π and $\mathcal{T}_K^\pi$, and the latter bootstraps from U_m itself.

For the optional contractive specialization, additionally assume the fixed-point map is measurable and U_* is bounded on the closure (the latter follows from Assumption A.3); set $U_*(s_{\text{abs}}) = -C_{\text{max}}$ and define

$$\varepsilon_{\text{res}}^* \triangleq \|U_* - \mathcal{T}_K^\pi U_*\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \quad (4)$$

Under Assumption 5.1 and the Lipschitz-head condition, consecutive latent differences satisfy $\|z^{(j+1)} - z^{(j)}\| \leq L_z^j C_z$ uniformly. Hence Theorem 6.1 gives the finite- m specialization

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq L_V C_z \frac{L_z^n (1 - L_z^{m-n})}{1 - L_z} + \frac{\|U_m - \mathcal{T}_K^\pi U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}}{1 - \gamma^K}. \quad (5)$$

Uniform convergence $U_m \rightarrow U_*$ and Bellman contraction imply convergence of the residual norm. Taking $m \rightarrow \infty$ recovers

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \underbrace{\frac{\varepsilon_{\text{res}}^*}{1 - \gamma^K}}_{\text{fixed-point residual}} + \underbrace{L_V \frac{L_z^n}{1 - L_z}}_{\text{geometric path bound}} C_z. \quad (6)$$

Thus the earlier fixed-point decomposition is a specialization obtained by geometrically bounding the recurrent path and sending the finite reference depth to infinity. It is not an assertion that arbitrary TRMs are designed to converge to a fixed point. Under $\|z_{\text{init}}\|_2 \leq R$ with $\mathcal{Z}_{\text{inv}} = B_R$, projection gives $C_z \leq 2R$ (Remark M.2).

A projection-active specialization makes the role of R explicit. When projection is active (that is, $\|f_\theta(z, y, x)\|_2 \geq \rho_R > R$ for all $z \in B_R$ and relevant (x, y)), Π_R also contracts the evaluator. Proposition M.3 gives

$$L_z \leq L_z^{\text{sat}}(R) := \frac{R}{\rho_R} L_z^{\text{pre}}(R), \quad (7)$$

and Proposition M.4 bounds the residual relative to a plan-independent baseline $b_0(x) = V_\psi(0, x)$:

$$\varepsilon_{\text{res}}^* \leq \varepsilon_0 + (1 + \gamma^K) L_V(R) R, \quad (8)$$

with $\varepsilon_0 := \|b_0 - \mathcal{T}_K^\pi b_0\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$. Projection thus enters both the architectural term (through $L_V(R)R$) and the truncation term (through $L_z^{\text{sat}}(R)$ and $C_z \leq 2R$). The experiments report finite-sample proxies for this regime. They do not certify the uniform ρ_R and Lipschitz assumptions; the held-out U_n Bellman residual upper-bounds $\varepsilon_{\text{res}}^*$ only up to the truncation gap (Remark K.3).

Corollary 6.2 (Finite-depth residual certificate). *Fix a policy π and a finite-depth evaluator U_n on the advantage-evaluation closure $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$. Define*

$$\varepsilon_{\text{res}, n} \triangleq \|U_n - \mathcal{T}_K^\pi U_n\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \quad (9)$$

Then

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{\varepsilon_{\text{res}, n}}{1 - \gamma^K}. \quad (10)$$

If $\hat{A} = \hat{A}_{U_n}$ is the exact one-step centered estimator of Eq. (67), then Assumption 6.4 holds with

$$\varepsilon_A \triangleq \frac{2\gamma \varepsilon_{\text{res}, n}}{1 - \gamma^K}. \quad (11)$$

Consequently, under the hypotheses of Theorem 6.5,

$$\eta(\pi_{\text{new}}) \geq \hat{L}_\pi(\pi_{\text{new}}) - \frac{2\alpha\gamma}{1 - \gamma} \frac{\varepsilon_{\text{res}, n}}{1 - \gamma^K} - \frac{2\varepsilon_{\text{CPI}}\gamma}{(1 - \gamma)^2} \alpha^2. \quad (12)$$

This certificate follows from Bellman contraction alone and applies in finite MDPs or deterministic edit environments where the K -step backup is computable. It does not require the Lipschitz or contractive architectural assumptions. A finite-MDP unit test checks 80 value/decomposition rows and 21 CPI rows with zero violations (Appendix C.1).

The CPI guarantees below are deterministic conditional statements that need sup-norm residual control on $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$. In finite or exactly evaluable MDPs Corollary 6.2 can certify it; otherwise it remains an assumption. Because the closure depends on both π and π_{cand} , residual control is a property of the policy pair. The certificate is post-candidate: once π_{cand} is chosen, the relevant closure can be checked; future candidates require their own check.

Corollary 6.3 (Advantage-error bridge). *Under the hypotheses of Theorem 6.1, define*

$$B_{n, m} := \|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + \frac{\|U_m - \mathcal{T}_K^\pi U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}}{1 - \gamma^K}. \quad (13)$$

Let $\hat{A} = \hat{A}_{U_n}$ be the exact one-step centered estimator of Eq. (67), with the transition expectation evaluated exactly (or applied to a deterministic transition) and the baseline computed exactly under π . Then Assumption 6.4 holds with the valid choice

$$\varepsilon_A = 2\gamma B_{n, m}. \quad (14)$$

If Assumption 5.1 and the L_V -Lipschitz head condition also hold, sending $m \rightarrow \infty$ gives the fixed-point specialization

$$\varepsilon_A \leq \underbrace{\frac{2\gamma \varepsilon_{\text{res}}^*}{1 - \gamma^K}}_{\text{architectural}} + \underbrace{2\gamma L_V \frac{L_z^n}{1 - L_z}}_{\text{unrolling}} C_z. \quad (15)$$

Here the value-error norm is taken over $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ (Definition I.2). For a centered K -step estimator, 2γ becomes $2\gamma^K$; GAE-style constants can likewise be absorbed into ε_A (Remark L.3). Combining the finite-reference choice with Theorem 6.5 gives the fixed-point-free penalty

$$\eta(\pi_{\text{new}}) \geq \hat{L}_\pi(\pi_{\text{new}}) - \frac{2\alpha\gamma}{1 - \gamma} B_{n,m} - \frac{2\varepsilon_{\text{CPI}}\gamma}{(1 - \gamma)^2} \alpha^2. \quad (16)$$

Thus a finite recurrent path and a residual at any deeper finite depth feed directly into CPI. Geometric L_z^n dependence is only the contractive specialization.

6.2 CONSERVATIVE POLICY IMPROVEMENT

For $\pi_{\text{new}} = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$, standard CPI gives monotonic improvement up to an $O(\alpha^2)$ penalty. The decompositions above enter through centered evaluation error: under exact centering, candidate-policy advantage error contributes the explicit linear term $\alpha \varepsilon_{A, \text{cand}} / (1 - \gamma)$.

Assumption 6.4 (Uniform advantage error on states visited by π). For some $\varepsilon_A \geq 0$:

$$|\hat{A}(s, a) - A^\pi(s, a)| \leq \varepsilon_A \quad (17)$$

for all (s, a) with $s \in \mathcal{R}_\pi$ and $a \in \text{supp}(\pi_{\text{cand}}(\cdot|s)) \cup \text{supp}(\pi(\cdot|s))$.

The relevant support depends on π_{cand} (standard CPI circularity; see Appendix I). Write $\hat{L}_\pi(\pi') := \eta(\pi) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\pi, a \sim \pi'(\cdot|s)} [\hat{A}(s, a)]$ for the CPI surrogate built from the approximate advantage.

Theorem 6.5 (CPI with centered evaluation error). *Let $\alpha \in [0, 1]$ and let $\pi_{\text{new}} = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$ be the exact mixture policy, with no distillation error. Suppose Assumption 6.4 holds. Assume the advantage estimator is statewise-centered under π :*

$$\mathbb{E}_{a \sim \pi(\cdot|s)} [\hat{A}(s, a)] = 0 \quad \text{for all } s \in \mathcal{R}_\pi. \quad (18)$$

We define the candidate-policy bias in advantage estimation

$$\varepsilon_{A, \text{cand}} \triangleq \sup_{s \in \mathcal{R}_\pi} \left| \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)} [\hat{A}(s, a) - A^\pi(s, a)] \right|. \quad (19)$$

Then:

$$\eta(\pi_{\text{new}}) \geq \hat{L}_\pi(\pi_{\text{new}}) - \frac{\alpha}{1 - \gamma} \varepsilon_{A, \text{cand}} - \frac{2\varepsilon_{\text{CPI}}\gamma}{(1 - \gamma)^2} \alpha^2, \quad (20)$$

where $\varepsilon_{\text{CPI}} \triangleq \sup_s |\mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)} A^\pi(s, a)|$.

The linear-in- α contribution $\alpha \varepsilon_{A, \text{cand}} / (1 - \gamma)$, instead of the naive $\varepsilon_A / (1 - \gamma)$, requires the exact-centering condition (18). That holds deterministically only when the baseline expectation in $\hat{A}$ is computed by exact summation over the action space, as in the discrete-action setting of Algorithm 1. For sampling-based or learned baselines the bound falls back to Corollary K.2.

The constants split into estimable and assumed (Table 3): $\hat{L}_z^{\text{post}}$, $\hat{\rho}_R$, $\hat{C}_z$, $\hat{L}_V$, and $\hat{\varepsilon}_{\text{cent}}$ admit finite-difference or batch proxies. The architectural residual $\varepsilon_{\text{res}}^*$ is a sup-norm over the advantage closure that we do not estimate uniformly; the held-out finite-batch residual upper-bounds it only up to the truncation gap (Remark K.3). On the theory-aligned episodic- z hard-suite rerun, the predicted CPI penalty is non-finite on 2/10 seeds at every α (Table 20). At this scale, the bound functions as a structural decomposition; its constants are not calibrated enough to predict Sudoku-scale improvement.

Corollary 6.6 (Sufficient condition for one-step improvement). *Under Theorem 6.5, if*

$$\hat{L}_\pi(\pi_{\text{new}}) - \eta(\pi) \geq \frac{\alpha}{1-\gamma} \varepsilon_{A,\text{cand}} + \frac{2\varepsilon_{\text{CPI}}\gamma}{(1-\gamma)^2} \alpha^2, \quad (21)$$

then $\eta(\pi_{\text{new}}) \geq \eta(\pi)$. In particular, under Corollary 6.3, it suffices to replace $\varepsilon_{A,\text{cand}}$ in (21) by $2\gamma B_{n,m}$. Under the optional contractive specialization, one may further upper-bound this quantity by

$$2\gamma \left(\frac{\varepsilon_{\text{res}}^*}{1-\gamma^K} + L_V \frac{L_z^n}{1-L_z} C_z \right).$$

Theorem 6.5 applies to the exact mixture; distillation error lies outside it. Exact baseline summation over a discrete action space satisfies (18); approximate centering replaces the linear term by $[(1-\alpha)\varepsilon_{\text{cent}} + \alpha\varepsilon_{A,\text{cand}}]/(1-\gamma)$ via Corollary K.2. The theorem is most informative in the small- α regime, where the linear term dominates the α^2 penalty.

Corollary 6.7 (One-step parameter dependence). *Under Corollary 6.3 and Theorem 6.5,*

$$\eta(\pi_{\text{new}}) \geq \hat{L}_\pi(\pi_{\text{new}}) - \frac{2\alpha\gamma}{1-\gamma} \left(\frac{\varepsilon_{\text{res}}^*}{1-\gamma^K} + L_V \frac{L_z^n}{1-L_z} C_z \right) - \frac{2\varepsilon_{\text{CPI}}\gamma}{(1-\gamma)^2} \alpha^2. \quad (22)$$

Corollary 6.7 is the optional fixed-point specialization of the finite-reference penalty $2\alpha\gamma B_{n,m}/(1-\gamma)$. It makes the one-step dependence on (L_z, n, K, α) explicit and reduces to standard CPI in the limit $L_z, \varepsilon_{\text{res}}^* \rightarrow 0$. Two constants in (22) are observed rather than tuned: L_V is only locally estimated by finite differences (Table 4), and the current clamping mechanism produces little range in the projection-active proxy $\hat{L}_z^{\text{post}}$ (≈ 0.22 – 0.24 ; Appendix D.7, Table 11). On the hard suite, we therefore treat the L_z -dependent terms as monitoring quantities.

A projection-aware version of (22) substitutes $\varepsilon_0 + (1+\gamma^K)L_V(R)R$ for $\varepsilon_{\text{res}}^*$ and $L_z^{\text{sat}}(R)$ for L_z ; the full statement is Eq. (70) in Appendix M.5. In the saturated regime ($\rho_R \gg R$), R/ρ_R sets $L_z^{\text{sat}}(R)$ more directly than the spectral-norm target does (Appendix D.7). Smaller L_z or larger n reduces L_z^n ; smaller α is safer; larger K tightens $1/(1-\gamma^K)$ at the cost of a K -dependent residual. Persistent- z uses the direct augmented-state residual of Appendix B; slow drift enters only the optional comparison to a memoryless fixed-point reference.

7 EXPERIMENTS

All task results in this paper are on 4×4 Sudoku. The decision-relevant evidence is a theory-aligned episodic run at 0%, a persistent distilled run at 57.4% under a different interaction budget, and a controlled projection $\times$ contraction factorial. These implementations do not validate the fixed-snapshot theorem; Appendix Table 2 maps each claim to its conditions and status.

Setup. The persistent result, factorial, and nominal baselines use no-mask 6–8-empty Sudoku ($T=16$) for 20k outer steps; the episodic run stops at 20k environment steps. Nominal interactions are unequal. Architecture-matched baselines share the TRM backbone but not the update rule; external SB3 baselines are implementation checks. Contraction uses operator-norm clamping at $L_z=0.9$; secondary diagnostics use the easier 1–4-empty suite.

7.1 PERSISTENT HARD-SUITE IMPLEMENTATION AND LIMITATIONS

The persistent, distilled implementation produces the higher task result in Table 1. The no-mask 6–8-empty protocol is the first setting in this paper where masked random play scores 0%.

At the nominal endpoint, the +25.4pp TRM+PPO gap is distinguishable in the retained analysis (Welch $t \approx 4.10$, $p \approx 7.3 \times 10^{-4}$, 95% CI [12.4, 38.5]pp; consistent bootstrap/permutation tests). This joint, unequal-interaction implementation contrast isolates no component and implies no efficiency advantage (Appendix G.2).

For the persistent run, clock/carry details are unaudited, the uniform augmented residual and centering are unmeasured, and distillation violates exact-mixture deployment. Thus 57.4% is an implementation result, not verification of Corollary B.1 or Theorem 6.5; exact centering alone would not repair the mixture mismatch.

Table 1: No-mask hard 4×4 results on 6–8 empties ($T=16$, mean $\pm$ std across seeds, nominal 20k outer-step protocol; persistent- z). The protocol does not equalize environment interactions across rows; see paragraph below.

Method	Outer Steps	Seeds	Success (%)	Return	Invalid
UPI-TRM	20k	10	57.4 ± 12.2	-1.82 ± 2.66	0.000 ± 0.000
<i>Architecture-matched (same TRM backbone, no invalid actions):</i>					
TRM+PPO	20k	10	32.0 ± 15.3	-7.87 ± 3.65	0.000 ± 0.000
TRM+A2C	20k	10	0.0 ± 0.0	-16.81 ± 0.29	0.000 ± 0.000
TRM+DQN	20k	10	0.0 ± 0.0	-18.38 ± 0.63	0.000 ± 0.000
<i>External (SB3, flat MLP policy; high invalid-action rates):</i>					
SB3 PPO/A2C/DQN/DQN($n=5$)	20k	10	0.0 ± 0.0	≈ -17.9	$0.78-1.00$

7.2 PROJECTION AND CONTRACTION ABLATION

The 2×2 factorial varies projection and contraction (Appendix Table 17). Projection’s composite +13.0pp contrast ($36.2\% \rightarrow 49.2\%$) mixes clipping and training dynamics. Contraction’s +2.2pp is inconclusive; the source-reported $p \approx 0.6$ is not independently auditable from retained per-seed/test artifacts. Contraction remains a conditional error-control mechanism, not a demonstrated performance dial.

7.3 THEORY-ALIGNED EPISODIC NEGATIVE RESULT

The episodic run uses reinitialization, exact baselines/targets, and the direct mixture, obtaining 0.0% at 20k environment steps over seeds 41–50. Its four joint changes and different budget preclude a controlled comparison, and its residual is not the persistent augmented residual (Appendix G.3).

7.4 INTERACTION-BUDGET AUDIT

PPO/A2C/DQN collect 64/32/4 transitions per nominal step. UPI-TRM is absent from the common-budget reruns, whose provenance is missing; there is no matched UPI-TRM-PPO result (Appendix G.1).

7.5 FINITE-MDP NUMERICAL UNIT TEST

An exact 64-state, 4-action unit test has zero violations in 80 value and 21 selected-slice CPI rows; it is not an empirical discovery or Cartesian sweep (Appendix C.1).

7.6 SECONDARY DIAGNOSTICS

The easier diagnostic shows depth stability, not a hard-suite effect; other sweeps remain exploratory (Appendices F.1 and D).

8 DISCUSSION, LIMITATIONS, AND CONCLUSION

The primary result certifies a transient depth- n evaluator against any finite $m > n$ through its recurrent path and depth- m residual. A fixed point is unnecessary; contraction is only a geometric specialization. Persistent state has the same direct residual certificate on the augmented MDP, while slow drift serves only a stronger memoryless-reference comparison.

Policy improvement additionally requires uniform closure control, exact centering, and pointwise exact-mixture deployment at a frozen snapshot; it certifies neither training nor distillation. The headline run is distilled and lacks a uniform augmented residual and verified centering, so exact centering alone would not make Theorem 6.5 apply.

Empirically: episodic 0%; PPO unmatched; projection composite; contraction inconclusive.

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A SUPPLEMENTARY MATERIAL

Assumption A.1 (Bounded rewards). Potential-shaped rewards are uniformly bounded: $|r(s, a, s')| \leq R_{\max}$ for all transitions.

$$\begin{aligned} \Pi_R(z) &= \arg \min_{u: \|u\|_2 \leq R} \|u - z\|_2 \\ &= \begin{cases} z, & \|z\|_2 \leq R, \\ R \frac{z}{\|z\|_2}, & \|z\|_2 > R. \end{cases} \end{aligned} \quad (23)$$

$$G^{(K)}(s_0) = \sum_{k=0}^{K-1} \gamma^k r_k + \gamma^K \bar{V}(s_K). \quad (24)$$

$$C_z := \sup_{(x,y)} \|\tilde{f}_\theta(z_{\text{init}}(x, y), y, x) - z_{\text{init}}(x, y)\| < \infty. \quad (25)$$

Proposition A.2 (Fixed point and finite-unrolling bias (Banach)). *Under Assumption 5.1, the map $T_{x,y}(z) = \tilde{f}_\theta(z, y, x)$ has a unique fixed point $z^*(x, y)$, with linear convergence*

$$\|z^{(t)} - z^*(x, y)\| \leq L_z^t \|z^{(0)} - z^*(x, y)\|. \quad (26)$$

Moreover, for $z^{(0)} = z_{\text{init}}(x, y)$,

$$\|z^{(n)} - z^*(x, y)\| \leq \frac{L_z^n}{1 - L_z} C_z. \quad (27)$$

$$|V_\psi(z, x) - V_\psi(z', x)| \leq L_V \|z - z'\|. \quad (28)$$

Assumption A.3 (Bounded value head on the invariant region). There exists $V_{\max} < \infty$ such that $|V_\psi(z, x)| \leq V_{\max}$ and $|V_{\bar{\psi}}(z, x)| \leq V_{\max}$ on the invariant region.

For $\pi_{\text{new}} = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$:

$$\eta(\pi_{\text{new}}) \geq L_\pi(\pi_{\text{new}}) - \frac{2\varepsilon_{\text{CPI}}\gamma}{(1 - \gamma)^2} \alpha^2, \quad (29)$$

where $\varepsilon_{\text{CPI}} = \max_s |\mathbb{E}_{a \sim \pi_{\text{cand}}} A^\pi(s, a)|$.

$$\pi_{\text{new}} = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}, \quad \alpha \in [0, 1]. \quad (30)$$

B PERSISTENT- z ON THE AUGMENTED MDP

B.1 DIRECT FINITE-DEPTH RESIDUAL CERTIFICATE

Fix $n, (\theta, \psi)$, the policy parameters, and all environment kernels. Any timeout or remaining edit budget that affects termination is included in y , and edit updates that clock; we suppress it in the notation. Define

$$h_n(x, y, z) := T_{x,y}^{\circ n}(z) = (\Pi_R \circ f_\theta)^{\circ n}(z, y, x), \quad \bar{s} = (x, y, z), \quad (31)$$

and the augmented transition by first computing h_n , sampling an action, and then setting

$$\bar{s}' = (x, \text{edit}(y, a; x), h_n(x, y, z)).$$

The current and candidate action kernels share this same frozen h_n and hence the same augmented MDP:

$$\bar{\pi}(a \mid \bar{s}) = \pi_\phi(a \mid y, h_n(x, y, z), x), \quad \bar{\pi}_{\text{cand}}(a \mid \bar{s}) = \pi_{\text{cand}}(a \mid y, h_n(x, y, z), x).$$

Changing the recurrent map between these two policies would change the transition kernel and is outside ordinary CPI. The initial law is

$$\bar{\rho}(dx, dy, dz) = \rho(dx, dy) \delta_{z_{\text{init}}(x, y)}(dz), \quad (32)$$

or, for stochastic initialization, $\bar{\rho}(dx, dy, dz) = \rho(dx, dy) \nu_0(dz | x, y)$ with the recorded kernel ν_0 .

Assume the edit/reward kernels, h_n , V_ψ , and both policy kernels are measurable; rewards and the evaluator are bounded; and the augmented advantage-evaluation closure is measurable, contains the one-step successors under both action supports, and is $\bar{\pi}$ -forward invariant almost surely. Define

$$\tilde{U}_n(\bar{s}) := V_\psi(h_n(x, y, z), x), \quad \tilde{U}_n(\bar{s}_{\text{abs}}) := -C_{\text{max}}, \quad (33)$$

and let $\bar{T}_K^{\bar{\pi}}$ denote the K -step Bellman operator of this frozen augmented MDP. Its direct residual is

$$\bar{\varepsilon}_{\text{res}, n} := \|\tilde{U}_n - \bar{T}_K^{\bar{\pi}} \tilde{U}_n\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}. \quad (34)$$

Here $\eta(\bar{\pi}')$ denotes the discounted return of any augmented policy $\bar{\pi}'$ from $\bar{\rho}$, and $d_{\bar{\pi}}$ is the discounted occupancy of the current policy. For an augmented advantage estimator $\hat{A}$, define

$$\hat{L}_{\bar{\pi}}(\bar{\pi}') := \eta(\bar{\pi}) + \frac{1}{1 - \gamma} \mathbb{E}_{\bar{s} \sim d_{\bar{\pi}}, a \sim \bar{\pi}'(\cdot | \bar{s})} [\hat{A}(\bar{s}, a)], \quad (35)$$

$$\bar{\varepsilon}_{\text{CPI}} := \sup_{\bar{s}} |\mathbb{E}_{a \sim \bar{\pi}_{\text{cand}}(\cdot | \bar{s})} \hat{A}^{\bar{\pi}}(\bar{s}, a)|. \quad (36)$$

Corollary B.1 (Direct persistent- z certificate on the augmented MDP). *Under the preceding frozen-MDP, measurability, boundedness, and closure assumptions,*

$$\|\tilde{U}_n - V^{\bar{\pi}}\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq \frac{\bar{\varepsilon}_{\text{res}, n}}{1 - \gamma^K}. \quad (37)$$

If $\hat{A}_{\tilde{U}_n}$ uses the exact transition expectation and exact baseline and is therefore centered pointwise under $\bar{\pi}$ on the full augmented state, then Assumption 6.4 holds on the augmented closure with the valid choice

$$\bar{\varepsilon}_A := \frac{2\gamma \bar{\varepsilon}_{\text{res}, n}}{1 - \gamma^K}. \quad (38)$$

If, additionally, the deployed policy is the pointwise exact mixture $\bar{\pi}_{\text{new}} = (1 - \alpha)\bar{\pi} + \alpha\bar{\pi}_{\text{cand}}$ with no parameter interpolation or distillation, Theorem 6.5 gives

$$\eta(\bar{\pi}_{\text{new}}) \geq \hat{L}_{\bar{\pi}}(\bar{\pi}_{\text{new}}) - \frac{2\alpha\gamma}{1 - \gamma} \frac{\bar{\varepsilon}_{\text{res}, n}}{1 - \gamma^K} - \frac{2\bar{\varepsilon}_{\text{CPI}}\gamma}{(1 - \gamma)^2} \alpha^2. \quad (39)$$

This direct result is Bellman contraction on the actual finite-depth augmented evaluator. It requires no latent fixed point, recurrent contraction, Lipschitz head, projection, or slow-drift assumption. The augmented residual differs from the episodic residual on (x, y) because both the policy and transition depend on the carried latent. We do not measure $\bar{\varepsilon}_{\text{res}, n}$ uniformly for the headline run. Exact augmented-state centering is also not established there, and the headline implementation uses distillation, so the exact-mixture condition is known not to hold.

Remark B.2 (Scope of the direct augmented-state certificate). Corollary B.1 is a result on $\bar{s} = (x, y, z)$, not on the plan marginal (x, y) . Its exact-centering condition averages actions conditional on the full $\bar{s}$, and its mixture identity must hold pointwise on the augmented closure. Finite-batch maxima are diagnostics, not uniform certificates. The retained implementation package does not contain the training repository needed to verify latent-carry order, clock encoding, or whether candidate construction freezes the recurrent map; these remain implementation-audit requirements rather than assumed facts.

B.2 OPTIONAL SLOW-DRIFT COMPARISON TO A FIXED-POINT REFERENCE

Slow drift is needed only for the stronger architectural comparison between the actual transient evaluator and the memoryless reference $\tilde{U}_*(x, y, z) := V_\psi(z^*(x, y), x)$, with $\tilde{U}_*(\bar{s}_{\text{abs}}) = -C_{\text{max}}$. For this subsection, additionally assume $z^*(x, y)$ is measurable, $\tilde{U}_*$ is bounded and measurable on the augmented closure, and the tracking conditions below hold uniformly over trajectories formed by one action supported by either policy followed by $\bar{\pi}$ dynamics. These are the one-deviation trajectories generating the augmented advantage closure. Let $e : \mathcal{Y} \rightarrow \mathcal{E}_y$ be a fixed plan embedding and write $\|y - y'\|_{\mathcal{E}} := \|e(y) - e(y')\|_{\mathcal{E}_y}$.

Assumption B.3 (Slow drift of fixed points). Uniformly on the augmented policy-pair closure, there exist $L_{z^*} \geq 0$, $\Delta y_{\max} \geq 0$, and $E_0 < \infty$ such that

1. $\|z^*(x, y) - z^*(x, y')\| \leq L_{z^*} \|y - y'\|_{\mathcal{E}}$;
2. every admissible edit satisfies $\|y_{t+1} - y_t\|_{\mathcal{E}} \leq \Delta y_{\max}$; and
3. the initial carried latent satisfies $\|z_0 - z^*(x, y_0)\| \leq E_0$.

Remark B.4 (A conditional sufficient bound for L_{z^*}). If y has a specified differentiable continuous representation, f_{θ} is C^1 near an interior fixed point, and $\|\partial_z f_{\theta}\|_{\text{op}} \leq L_z < 1$, the implicit-function calculation gives

$$L_{z^*} \leq \frac{\|\partial_y f_{\theta}\|_{\text{op}}}{1 - L_z}. \quad (40)$$

For discrete Sudoku plans this calculation additionally requires a differentiable extension off the discrete embedding; the manuscript does not provide or verify such an extension.

Remark B.5 (Discrete-plan scope). For a discrete domain, Assumption B.3 is simply an additional regularity assumption in the stated embedding. Successful training does not by itself establish it. We do not verify it for the headline Sudoku run.

Remark B.6 (Scope of the slow-drift assumption). Only the augmented closure for the frozen policy pair is relevant; no global smoothness claim outside that closure is needed.

Define $e_t = \|z_t - z^*(x, y_t)\|$ and $\kappa_n = L_z^n$.

Lemma B.7 (Drift bound for persistent- z). Under Assumptions 5.1 and B.3,

$$e_{t+1} \leq \kappa_n e_t + L_{z^*} \Delta y_{\max}, \quad (41)$$

and

$$\sup_t \|z_t^{(n)} - z^*(x, y_t)\| \leq C_{\text{drift}}(n) := \kappa_n \max\left(E_0, \frac{L_{z^*} \Delta y_{\max}}{1 - \kappa_n}\right) \leq \kappa_n E_0 + \frac{\kappa_n L_{z^*} \Delta y_{\max}}{1 - \kappa_n}. \quad (42)$$

Remark B.8 (When the drift bound is informative). The bound is useful only when its assumed constants are finite and calibrated on the policy-pair closure. Otherwise the direct residual certificate remains mathematically valid but uncalibrated, while the fixed-point comparison is vacuous.

Define

$$\bar{\varepsilon}_{\text{res},*} := \|\tilde{U}_* - \bar{T}_K^{\bar{\pi}} \tilde{U}_*\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}. \quad (43)$$

Then Lipschitzness of the head and Lemma B.7 give

$$\|\tilde{U}_n - \tilde{U}_*\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq L_V C_{\text{drift}}(n), \quad (44)$$

$$\|\tilde{U}_n - V^{\bar{\pi}}\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq L_V C_{\text{drift}}(n) + \frac{\bar{\varepsilon}_{\text{res},*}}{1 - \gamma^K}, \quad (45)$$

$$|\bar{\varepsilon}_{\text{res},n} - \bar{\varepsilon}_{\text{res},*}| \leq (1 + \gamma^K) L_V C_{\text{drift}}(n). \quad (46)$$

These are optional comparisons to a memoryless fixed-point reference; none is needed for Corollary B.1.

C DIAGNOSTICS AND PRACTICAL PROXIES

Table 3 shows the theoretical bound quantities and the practical proxies typically logged during training.

C.1 FINITE-MDP CERTIFICATE

To check that the residual and CPI inequalities hold as stated in a setting where every quantity is computable, we run a controlled finite MDP. This is not a Sudoku benchmark; it checks the certificate mechanism on a 64-state, 4-action discounted MDP with bounded rewards in $[-1, 1]$ and all quantities $(V^{\pi}, Q^{\pi}, A^{\pi}, d_{\pi}, \eta(\pi))$ solved by matrix inversion. We define a scalar contractive evaluator by

$$z_{t+1}(s) = U_*(s) + L_z(z_t(s) - U_*(s)), \quad U_*(s) = V^{\pi}(s) + \xi(s), \quad U_n = z_n,$$

Table 2: Claim status. “Measured” denotes retained finite-sample evidence, not a uniform certificate.

Claim	Required condition	Evidence / status
Finite-depth value	Frozen policy pair, common closure and norm	Proven; Theorem 6.1
Persistent direct bound	Complete augmented state, shared frozen map, uniform residual	Valid but uncalibrated; residual not measured uniformly
CPI headline	Exact centering and exact deployed mixture	Not covered; headline policy is distilled
Persistent 57.4%	Nominal implementation protocol	10-seed implementation result; unequal interactions
Projection / contraction	Controlled factorial	+13.0pp composite / +2.2pp inconclusive
Finite-MDP inequalities	Exact finite-state calculation	Numerical unit test; 80 value + 21 CPI rows

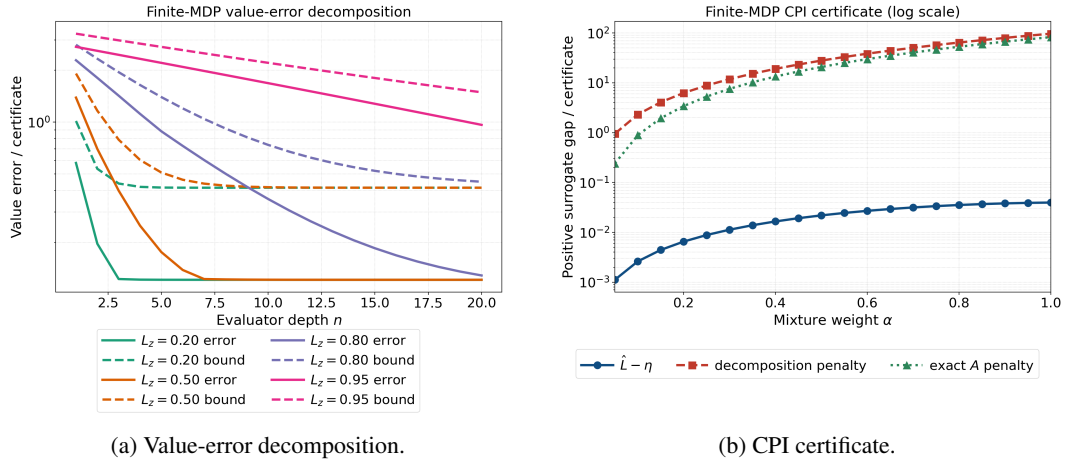


Figure 2: Finite-MDP certificate diagnostic ($|\mathcal{S}|=64$, $|\mathcal{A}|=4$). Left: value error $\|U_n - V^\pi\|_\infty$ decays with evaluator depth n and stays below the decomposition bound; larger L_z slows the decay. Right: for $\pi_\alpha = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$, the surrogate gap (blue) stays below both the decomposition-based CPI penalty (red) and the tighter $\varepsilon_{A,\text{cand}}$ penalty (green); log scale. All quantities are computed exactly by finite-state dynamic programming.

with $z_0(s) = 0$, so both the truncation term and the K -step residuals are known in closed form. From U_n we form the centered estimator $\hat{A}_{U_n}$, set $\pi_{\text{cand}}(\cdot | s) = \text{softmax}(\beta \hat{A}_{U_n}(s, \cdot))$, and compute $\eta(\pi_\alpha)$ and $\hat{L}_\pi(\pi_\alpha)$ for $\pi_\alpha = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$ via matrix inversion.

Figure 2 shows two checks. Left: for $\gamma=0.9$, $K=3$, $\sigma_\xi=0.05$, the value error decays with n and worsens as $L_z \rightarrow 1$, staying below the contractive specialization of Theorem 6.1. Right: on an evaluator-based candidate slice ($L_z=0.2$, $n=16$, $\sigma_\xi=0.1$, $\beta=1$), the surrogate gap $\hat{L}_\pi(\pi_\alpha) - \eta(\pi_\alpha)$ is positive and stays below both the decomposition-based CPI penalty and the tighter $\varepsilon_{A,\text{cand}}$ penalty for every tested α (log scale; the bound is conservative). Corollary 6.2 is directly checkable here because $\|U_n - \mathcal{T}_K^\pi U_n\|_\infty$ is computable. The script checks 80 value/decomposition rows (4 L_z values by 20 depths) and 21 α rows on this selected CPI slice, with zero violations; it does not constitute a full Cartesian sweep over $(L_z, n, K, \sigma_\xi, \alpha)$.

For this revision we reran `python3experiments/finite_mdp_certificate.py--outdirresults/finite_mdp_certificate_validation`. The retained CSV contains 80 `value_curve` and 21 `cpi_curve` rows, with both certificate flags true in every row; stdout is retained in `reports/finite_mdp_certificate_validation.log`.

Table 3: Theory quantities (used in bounds) and practical diagnostics/proxies typically logged during training.

Quantity	Appears in	Practical proxy / diagnostic
L_z^{pre}	Assump. 5.1	Finite-diff Lipschitz of f_θ before projection
L_z^{post}	Assump. 5.1	Finite-diff Lipschitz of $f_\theta = \Pi_R \circ f_\theta$ after projection
$L_z^{\text{sat}}(R)$	Prop. M.3	$(R/\rho_R) L_z^{\text{pre}}(R)$; effective modulus in the saturated regime
ρ_R	Prop. M.3	min-batch $\ f_\theta(z, y, x)\ _2$ for $z \in B_R$; saturation margin
projection_active	App. D.7	Fraction of samples where $\ f_\theta(z)\ > R$ (clipped)
C_z	Assump. 5.1	Empirical $\ f_\theta(z_{\text{init}}, y, x) - z_{\text{init}}\ $ on batches
$L_V(R)$	Eq. (28), Prop. M.4	Lipschitz of V_ψ on B_R ; finite-diff check (Table 4)
ε_0	Prop. M.4	K -step TD error of zero-latent baseline $V_\psi(0, x)$
$\ U_n - \mathcal{T}_K^\pi U_n\ $	Proxy only	Held-out K -step TD error; proxy for $\varepsilon_{\text{res}}^*$ up to $O(\ U_n - U_*\)$
$\bar{\varepsilon}_{\text{res}, n}$	Cor. B.1	Finite-batch augmented residual; not measured uniformly for head-line
$\varepsilon_{\text{cent}}$	Thm. 6.5	$\sup_s \mathbb{E}_{a \sim \pi} \hat{A}(s, a) $ (exactly 0 if baseline is exact)
$\varepsilon_{A, \text{cand}}$	Thm. 6.5	Empirical bias of $\mathbb{E}_{a \sim \pi_{\text{cand}}} [\hat{A}]$ on replay states
$\text{KL}(\pi_{\text{new}} \parallel \pi_\phi)$	App. N	Distillation projection gap measured on replay states

Table 4: Finite-difference $\hat{L}_V$ estimates for the theorem-facing Exp1 refreeze isolation package on the frozen B0 batch at $n_{\text{eval}}=8$. Each entry is mean $\pm$ sample-std over 10 seeds of the per-seed average across 8 repeated random-direction trials (16 perturbations per trial).

Condition	ϵ	$\hat{L}_V$	Seed range
Model A' (no contraction)	1e-05	0.192 $\pm$ 0.050	[0.143, 0.274]
Model A' (no contraction)	3e-05	0.111 $\pm$ 0.012	[0.095, 0.127]
Model A' (no contraction)	1e-04	0.095 $\pm$ 0.009	[0.082, 0.106]
Model A' (no contraction)	3e-04	0.096 $\pm$ 0.011	[0.077, 0.110]
Model A' (no contraction)	1e-03	0.097 $\pm$ 0.008	[0.080, 0.107]
Model B (contraction)	1e-05	0.200 $\pm$ 0.033	[0.143, 0.250]
Model B (contraction)	3e-05	0.108 $\pm$ 0.023	[0.075, 0.139]
Model B (contraction)	1e-04	0.090 $\pm$ 0.025	[0.056, 0.135]
Model B (contraction)	3e-04	0.087 $\pm$ 0.025	[0.051, 0.122]
Model B (contraction)	1e-03	0.085 $\pm$ 0.026	[0.050, 0.131]

D SECONDARY AND EXPLORATORY DIAGNOSTICS

The following studies probe mechanisms and implementation behavior. They are not theorem validation and are not used as confirmatory hard-suite evidence unless explicitly stated.

Finite-difference L_V diagnostic for the main comparison. To partially operationalize L_V in the same regime used by the main bound, we measure a local finite-difference proxy on the Exp1 refreeze package: frozen B0 states, episodic- z , $n_{\text{eval}}=8$, and seeds 41–50. Table 4 reports the seed-aggregated estimates. The smallest perturbation scale $\epsilon = 10^{-5}$ is noisier, but over the practical range $\epsilon \in \{3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$ both conditions stay in the same-order 0.085–0.111 band. This remains a local numerical diagnostic rather than a certified global Lipschitz bound, but it reduces the concern that the comparison is dominated by orders-of-magnitude differences in value-head sensitivity.

D.1 SPECTRAL NORMALIZATION INSTABILITY AND A MORE ROBUST ALTERNATIVE

The contraction assumption (Assumption 5.1) is a property of the latent update map $z \mapsto \tilde{f}_\theta(z, y, x)$. We first tried to enforce it by applying power-iteration spectral normalization (Miyato et al., 2018) to linear layers in the inner recursion, followed by a strict scalar contraction factor. On our TRM inner module this turned out to be numerically unstable.

Table 5: Representative local Lipschitz proxy $\hat{L}_z$ for the inner latent update map under different “contraction enforcement” components (4×4 Sudoku batch; $\varepsilon = 10^{-3}$). Power-iteration spectral normalization (SN) can cause the proxy to explode in our inner recursion, whereas operator-norm clamping (CLAMP) remains stable.

Variant	$\hat{L}_z$ (proxy)	Status
OFF	≈ 0.86	stable
SN-only	$\approx 2.7 \times 10^4$	exploded
SCALE-only	≈ 0.85	stable
SN+SCALE	$\approx 4.5 \times 10^4$	exploded
CLAMP-only	≈ 0.6	stable
CLAMP+SCALE	≈ 0.6	stable

Table 6: Latent collapse diagnostic at $n=16$ unroll depth. Comparing B vs. A isolates $z \rightarrow z$ contraction effect; comparing C vs. A isolates value-head normalization effect. Pre-projection metrics are measured before the ball projection Π_R ; post-projection norms are $\|z_{\text{post}}^{(n)}\| = \|\Pi_R(z_{\text{pre}}^{(n)})\|$.

Cond.	$z \rightarrow z$	V-head	$\ z_{\text{pre}}^{(n)}\ $	$\text{Var}(z_{\text{pre}}^{(n)})$	CosSim	$\text{Var}(V)$	$\ z_{\text{post}}^{(n)}\ $
A	OFF	OFF	32.0	814.0	0.196	0.016	10.0
B	ON	OFF	32.0	608.7	0.403	0.016	10.0
C	OFF	ON	32.0	814.0	0.200	2493	10.0
D	ON	ON	32.0	608.7	0.396	2412	10.0

We estimate a local Lipschitz proxy $\hat{L}_z$ by finite differences: sample a small perturbation δ with $\|\delta\|_2 = \varepsilon$ and compute $\|\tilde{f}_\theta(z + \delta, y, x) - \tilde{f}_\theta(z, y, x)\|_2 / \varepsilon$, averaging over random draws. We compare six variants: (i) OFF (no normalization, no scaling), (ii) SN-only, (iii) SCALE-only (only explicit per-layer output scaling), (iv) SN+SCALE, (v) CLAMP-only (operator-norm clamping without scaling), and (vi) CLAMP+SCALE.

Naively applying power-iteration spectral normalization to this architecture does not give reliable contraction control of the inner recursion. Explicit operator-norm clamping, by contrast, stays stable in this check and provides a practical path toward enforcing Assumption 5.1 without numerical blow-ups.

D.2 LATENT COLLAPSE: ISOLATING $z \rightarrow z$ CONTRACTION FROM VALUE-HEAD NORMALIZATION

A natural hypothesis for why $z \rightarrow z$ contraction enforcement can hurt success rates is *latent collapse*: contraction might drive all latent representations toward zero and destroy discriminative information. We test this with a controlled 2×2 that isolates $z \rightarrow z$ contraction from value-head normalization, which also affects $\text{Var}(V)$ but is architecturally separate.

The setup is a 2×2 factorial on the 4×4 Sudoku dataset (1–4 empties, single seed, unroll depth $n=16$, batch size 128, projection radius $R=10$):

- **Axis 1 ($z \rightarrow z$ contraction):** OFF (baseline) vs. ON (target $L_z = 0.9$).
- **Axis 2 (value-head normalization):** OFF (baseline) vs. ON (target $L_V = 1.0$).

This gives four conditions: A (both OFF), B ($z \rightarrow z$ ON, value-head OFF), C ($z \rightarrow z$ OFF, value-head ON), D (both ON). We measure pre-projection latent norms $\|z_{\text{pre}}^{(n)}\|$ and total variance $\text{Var}(z_{\text{pre}}^{(n)})$, pairwise cosine similarity, value variance $\text{Var}(V)$, and post-projection norms $\|z_{\text{post}}^{(n)}\| = \|\Pi_R(z_{\text{pre}}^{(n)})\|$.

Three observations stand out. First, $z \rightarrow z$ contraction causes modest clustering, not collapse-to-zero. Comparing B vs. A, total variance decreases by $\sim 25\%$ (ratio 0.75) and pairwise cosine similarity increases by ~ 0.21 , while pre-projection norms remain $\|z_{\text{pre}}^{(n)}\| \approx 32$, well above the projection radius $R=10$: latents become modestly more clustered in direction, but do not collapse toward zero. Second, the large $\text{Var}(V)$ differences come from value-head normalization, not $z \rightarrow z$ contraction. Comparing

Table 7: Toy-suite feasibility (4×4 Sudoku, 1–4 empties): success rate after 5000 training steps. Mean $\pm$ std over 3 seeds.

Method	Success (%)
UPI-TRM (persistent- z , no contraction, $R=10$)	93.3 ± 2.3
UPI-TRM (episodic- z , no contraction, $R=10$)	93.3 ± 2.3
UPI-TRM (episodic- z , contraction, proj. off)	90.7 ± 5.0
Random (masked uniform)	52.0

Table 8: Unroll sensitivity on B0 (initial states). Training depth $n_{\text{train}}=2$, evaluated at fixed $4 \times$ depth ($n_{\text{eval}}=8$).

Condition	Δ_V	Δ_π	Δ_z	Argmax
No Contraction	0.166 ± 0.287	0.0002 ± 0.0006	4.26 ± 1.53	$0.902_{[0.88, 0.92]}$
Contraction	0.064 ± 0.060	0.0001 ± 0.0002	1.67 ± 0.28	$0.955_{[0.94, 0.97]}$

C vs. A, $\text{Var}(V)$ increases by $\sim 160,000 \times$ (from 0.016 to 2493) when value-head normalization is enabled, while B vs. A barely moves (ratio ≈ 1.04); earlier observations of exploding $\text{Var}(V)$ “with contraction ON” were confounded by value-head normalization. Third, projection saturation explains the constant post-projection norms: all conditions have pre-projection $\|z_{\text{pre}}^{(n)}\| \approx 32 \gg R=10$, so Π_R is always active and clips to $\|z_{\text{post}}^{(n)}\| = 10$.

Within this single-seed diagnostic, the modest clustering pattern is inconsistent with a simple collapse-to-zero explanation, but it cannot identify the mechanism behind success-rate differences.

A 200-step follow-up associates value-head normalization with clamp-hitting: contraction alone remains stable with targets ≈ -1.7 , value-head normalization alone reaches the ± 20 clamps, and the combined cell reaches the positive clamp (target = 20 by step 60, $V(s) = 20$ by step 100). Short contraction-only runs with value-head normalization disabled remain stable but obtain only $\approx 10\%$ success at 1k steps. These exploratory observations identify a confound; they do not establish a causal or general training mechanism.

D.3 TOY-SUITE SANITY CHECK

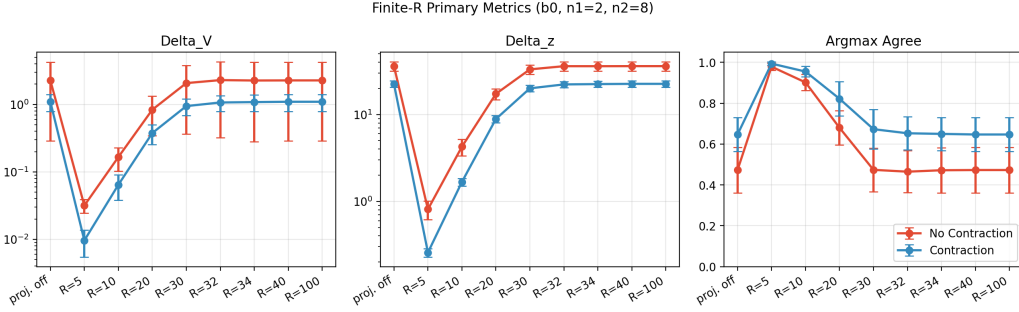
The 1–4-empty toy suite is a sanity check because masked random already succeeds 52% of the time; the depth-mismatch results reported on it (Appendix F.1) are calibration only. The full success-rate table is recorded here for completeness.

D.4 ISOLATION STUDY: DETAILED TABLES

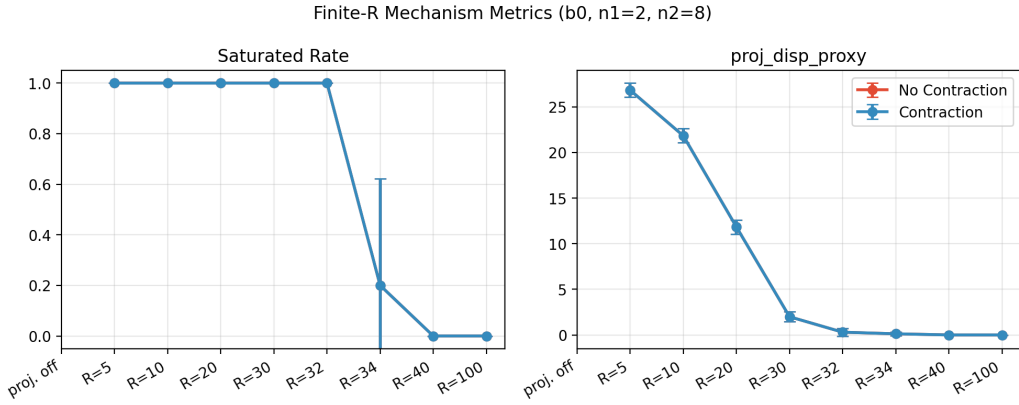
Tables 8 and 9 provide the full numeric results for the B0 isolation study (Section F.1).

D.5 FINITE- R SWEEP ON B0 (FIXED PAIR)

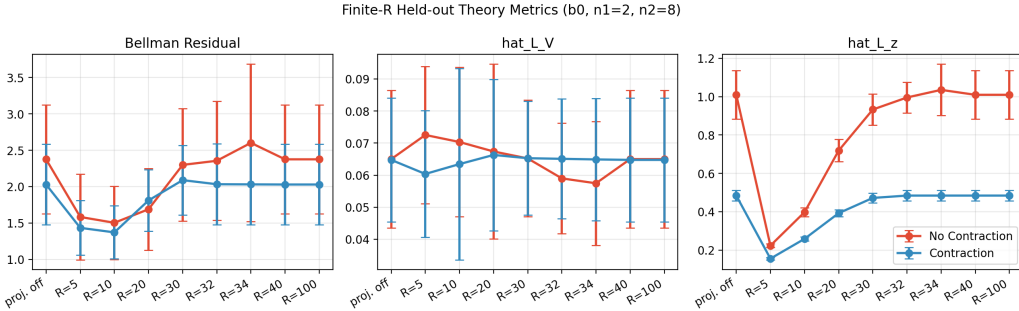
To test whether the finite- R specialization in Remark M.2 has empirical contact with the evaluator-truncation channel, we reran the fixed B0 pair $(n_{\text{train}}, n_{\text{eval}}) = (2, 8)$ from Section F.1 over radii $\{\text{proj. off}, 5, 10, 20, 30, 32, 34, 40, 100\}$ and aggregated seed-level means over seeds 41–50. The fixed-pair analysis passes the endpoint-separation and inactive-plateau checks, the plateau is recovered exactly by $R \geq 40$, and the source pre-projection norm at $R=10$ lies in a narrow p_{05} – p_{95} band of about 30.6–33.2, which is why the informative transition window is narrow. Across both models, $R=20$ is the cleanest shared intermediate regime. Adding held-out theory summaries shows that the measurable Bellman residual moves in that same narrow window, but non-monotonically in R : it drops from the inactive plateau to a low point around $R=10$, rises again through the intermediate window, and returns to the inactive plateau for $R \geq 40$. This gives finite- R contact with the residual channel as well as the truncation channel, while still not identifying $\varepsilon_{\text{res}}^*(R)$ end-to-end.



(a) Primary stability metrics.



(b) Mechanism diagnostics.



(c) Held-out theory diagnostics.

Figure 3: Fixed-pair finite- R sweep on B0 initial states with $n_{\text{train}}=2$, $n_{\text{eval}}=8$, and seeds 41–50. Top: the primary metrics (Δ_V , Δ_z , argmax agreement, Δ_π) move smoothly between the active-projection regime and the inactive plateau, with $R=20$ as the cleanest shared intermediate exhibit and $R \geq 40$ matching *proj. off*. Middle: the mechanism diagnostics show why this is not a gradual saturation knob. The saturated rate stays at 1.0 through $R \leq 32$, drops to 0.2 at $R=34$, and to 0.0 at $R \geq 40$, while *proj_disp_proxy* is only a scalar proxy based on norms, not a displacement norm. Bottom: held-out theory diagnostics show that the measurable Bellman residual also moves in the same transition window, but non-monotonically in R , while $\hat{L}_V$ remains same-order across radii. This appendix result therefore provides bounded empirical contact for both the truncation channel and the measurable residual channel, but it still does not identify any R -dependence of $\varepsilon_{\text{res}}^*$ end-to-end or explain projection dominance on the hard suite.

Table 9: Projection radius sweep on B0 (initial states), using the fixed $n = 2 \rightarrow 8$ mismatch. Here *proj. off* denotes the identity-map setting $\Pi_R = \text{Id}$; contraction still provides $2.1\times$ improvement.

Radius	Condition	Δ_V	Δ_z	Argmax
proj. off	No Contraction	2.275 ± 3.413	36.1 ± 7.9	0.47 [0.44,0.50]
	Contraction	1.097 ± 0.912	22.6 ± 3.3	0.65 [0.62,0.68]
R=10	No Contraction	0.166 ± 0.287	4.3 ± 1.5	0.90 [0.88,0.92]
	Contraction	0.064 ± 0.060	1.7 ± 0.3	0.95 [0.94,0.97]
R=100	No Contraction	2.275 ± 3.413	36.1 ± 7.9	0.47 [0.44,0.50]
	Contraction	1.097 ± 0.912	22.6 ± 3.3	0.65 [0.62,0.68]

Note: The $R = 100$ rows match the *proj. off* rows because pre-projection latent norms stay well below 100, so clipping never activates at this radius.

D.6 ISOLATION STUDY: B1 SUCCESSOR STATES

We present the full B1 (one-step successor states from B0) results from the isolation study (Section F.1).

Unroll sensitivity on B1. On successor states, contraction still improves Δ_V , Δ_z , Δ_π , and action agreement at all tested depth mismatches, though absolute instability remains higher than on B0 initial states (Figure 4).

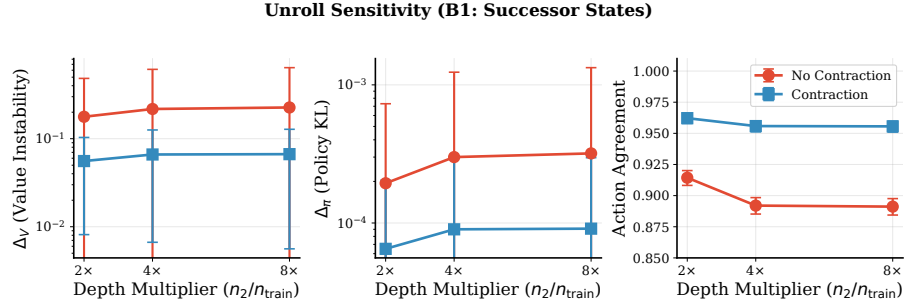


Figure 4: Unroll sensitivity on B1 (successor states). Training depth $n_{\text{train}}=2$, evaluated at $n_{\text{eval}} \in \{4, 8, 16\}$. “No Contraction” vs. “Contraction” ($L_z=0.9$); the *only* difference is `enable_contraction`. Both conditions: value-head spectral norm disabled, projection $R=10$. Seeds: 41–50. Left: Δ_V (value instability, log scale). Center: Δ_π (policy KL, log scale). Right: Action Agreement. Contraction improves all three metrics on B1 as well, though the absolute variability remains higher than on B0. Error bars: mean $\pm$ std (Δ_V , Δ_π); Wilson 95% CI (argmax).

Radius sweep on B1. On B1 at *proj. off*, contraction improves latent drift (Δ_z : $35.3 \rightarrow 22.7$, i.e., $1.6\times$), action agreement ($47.2\% \rightarrow 64.7\%$), and value instability Δ_V ($2.23 \rightarrow 1.06$, i.e., $2.1\times$). Absolute variability nevertheless remains substantially higher than on B0 initial states.

The main-text claim in Section F.1 stays anchored on B0 initial states, because B1 is an appendix-only successor-state check and is noisier even though the qualitative direction now matches B0.

D.7 PROJECTION DOMINANCE STUDY

We probed a putative “contraction dial” by sweeping the target $L_z^{\text{target}} \in \{0.9, 0.95, 0.99, 0.999\}$. All models are trained on the same 4×4 Sudoku suite (1–4 empties) with `disable_value_head_norm: true`, evaluated at $8\times$ depth mismatch on B0 (initial states).

The dial does not control achieved L_z . Figure 6 and Table 11 show that the achieved post-projection contraction modulus $\hat{L}_z^{\text{post}}$ stays in a narrow 0.219–0.242 band across the four dial settings, and the *ordering* of dial values does not produce a monotonic ordering in stability metrics. The dial has insufficient range to meaningfully control the achieved Lipschitz constant.

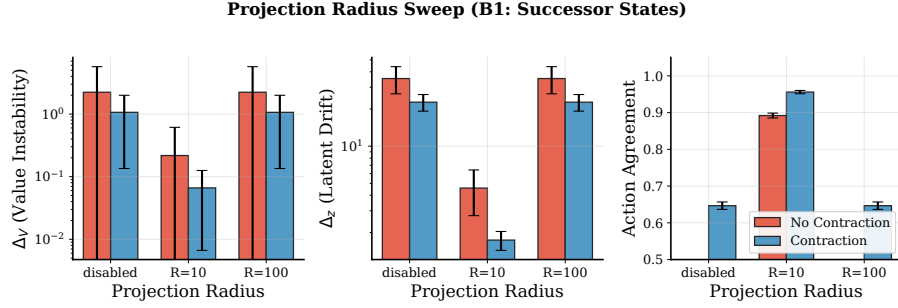


Figure 5: Projection radius sweep on B1 (successor states). “No Contraction” vs. “Contraction” ($L_z=0.9$) across projection settings {proj. off, $R=10$, $R=100$ }, where *proj. off* denotes $\Pi_R = \text{Id}$. Both conditions: value-head spectral norm disabled. Left: Δ_V . Center: Δ_z . Right: argmax agreement. At *proj. off*, contraction improves Δ_V , Δ_z , and argmax, though the absolute instability remains higher than on B0. Seeds: 41–50. Error bars: mean $\pm$ std (Δ_V , Δ_z); Wilson 95% CI (argmax).

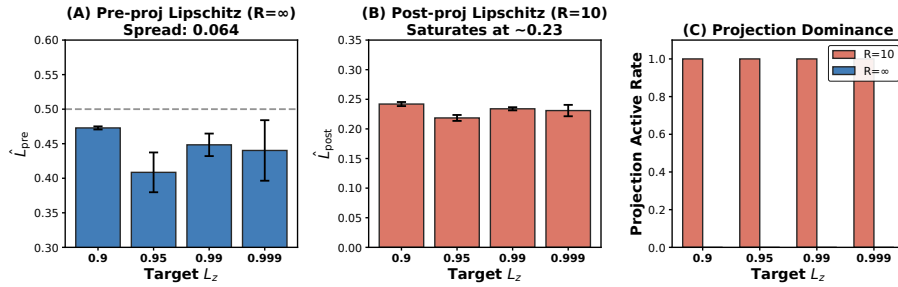


Figure 6: The contraction “dial” (L_z^{target}) does not meaningfully control achieved $\hat{L}_z^{\text{post}}$: all four settings land in a narrow 0.219–0.242 band. The ordering of dial values does not produce monotonic stability. All conditions: value-head spectral norm disabled, projection $R=10$. *No monotonic stability-dial claim is supported.*

Table 10: Radius sweep on B1 (successor states). Under the refrozen 10-seed protocol, contraction improves all three metrics with *proj. off*, though absolute instability remains higher than on B0.

Radius	Condition	Δ_V	Δ_z	Argmax
proj. off	No Contraction	2.230 ± 3.477	35.3 ± 8.7	0.47 [0.46,0.48]
	Contraction	1.064 ± 0.929	22.7 ± 3.5	0.65 [0.64,0.66]
R=10	No Contraction	0.218 ± 0.396	4.6 ± 1.8	0.89 [0.89,0.90]
	Contraction	0.066 ± 0.060	1.7 ± 0.3	0.96 [0.95,0.96]
R=100	No Contraction	2.230 ± 3.477	35.3 ± 8.7	0.47 [0.46,0.48]
	Contraction	1.064 ± 0.929	22.7 ± 3.5	0.65 [0.64,0.66]

Note: The $R = 100$ rows match the *proj. off* rows because pre-projection latent norms stay well below 100, so clipping never activates at this radius.

Table 11: Spectral-norm dial does not control Lipschitz constant. At $R=10$, projection dominates (100% active) and $\hat{L}_{post}$ saturates. With *proj. off* ($\Pi_R = \text{Id}$), $\hat{L}_{pre}$ varies weakly (spread 0.064 < 0.08 threshold) and non-monotonically.

Target L_z	$\hat{L}_{pre}$ (<i>proj. off</i>)	$\hat{L}_{post}$ ($R=10$)	Proj Active ($R=10$)	Proj Active (<i>proj. off</i>)
0.9	0.473 ± 0.002	0.242 ± 0.003	100%	0%
0.95	0.409 ± 0.029	0.219 ± 0.005	100%	0%
0.99	0.448 ± 0.016	0.234 ± 0.003	100%	0%
0.999	0.440 ± 0.044	0.231 ± 0.010	100%	0%
Spread	0.064	—	—	—

Projection, separately, is a major stabilizer. Figure 7 and Table 12 show that projection at $R=10$ improves Δ_V by 6–10 $\times$ and argmax agreement by ≈ 8 –10 pp compared to *proj. off*. At $R=10$ projection is active 100% of the time, which can mask any contraction-related effects.

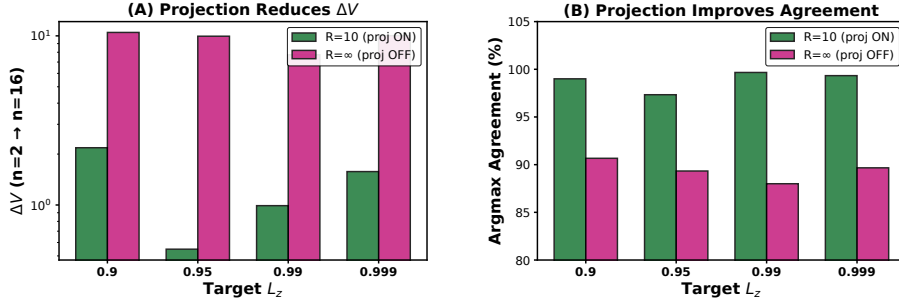


Figure 7: Projection ($R=10$ vs. *proj. off*, where *proj. off* denotes $\Pi_R = \text{Id}$) is a major stabilizer: Δ_V improves 6–10 $\times$; argmax agreement improves 8–10 pp. All conditions: value-head spectral norm disabled, $z \rightarrow z$ contraction ON ($L_z=0.9$). When projection is active 100%, it can dominate stability effects.

In this projection-active regime, the spectral-norm target does not meaningfully vary the achieved contraction modulus, while projection itself acts as a strong stabilizer. The isolation study therefore supports only a *binary* contraction effect at *proj. off*, not a tunable post-projection control parameter.

D.8 PROJECTION-FREE DIAL RANGE TEST

Appendix D.7 shows that the spectral-norm target L_z^{target} is not an effective control parameter when projection is active. We therefore test a narrower question here: with projection disabled, can inference-time scaling induce a wider range of achieved contraction?

With projection disabled (*proj. off*, implemented as $\Pi_R = \text{Id}$) and value-head spectral norm disabled, we apply inference-time scalar scaling to the $z \rightarrow z$ update weights of a single trained checkpoint,

Table 12: Projection is the primary stabilizer. $R=10$ projection dramatically improves stability metrics compared to *proj. off* ($\Pi_R = \text{Id}$).

Target L_z	ΔV (n=2→16)		Argmax Agree (%)	
	$R=10$	<i>proj. off</i>	$R=10$	<i>proj. off</i>
0.9	2.18	10.48	99.0	90.7
0.95	0.55	9.95	97.3	89.3
0.99	0.99	7.72	99.7	88.0
0.999	1.58	10.02	99.3	89.7
Mean	1.32	9.54	98.8	89.4
Range	[0.5, 2.2]	[7.7, 10.5]	[97, 100]	[88, 91]

Table 13: Projection-free dial range test via inference-time scaling. Single checkpoint with *proj. off*, value-head spectral normalization disabled, 100 eval samples at $8\times$ depth mismatch. The achieved L_z^{pre} proxy spans $\approx 0.52\text{--}0.95$.

Scale	L_z^{pre}	$\Delta_V @ 8\times$	Argmax@ $8\times$
1.00	0.9483	0.5663	0.6600
0.90	0.7192	0.4918	0.6600
0.80	0.5876	0.4174	0.7000
0.70	0.5195	0.4389	0.7400
0.60	0.5212	0.6385	0.6900

then measure the achieved pre-projection Lipschitz proxy L_z^{pre} and stability metrics (Δ_V , argmax agreement) at $8\times$ depth mismatch on 100 B0 samples.

The achieved L_z^{pre} span is wide ($\approx 0.52\text{--}0.95$; spread ≈ 0.43), so inference-time scaling *can* vary the proxy once projection is disabled. What is not supported is a *monotonic* stability relation: neither Δ_V nor argmax agreement changes monotonically with L_z^{pre} in this range. We therefore treat the run as a feasibility check only.

D.9 TRAINING STABILITY ABLATION

We ablate projection at *training time* to test whether the ball projection Π_R is necessary for stable training, as opposed to its role in evaluation-time mismatch stability (see projection dominance study above).

In this easy-suite ablation, all six conditions (contraction ON/OFF crossed with {*proj. off*, 10, 100}) pass the stated stability gate and obtain 94–98% success. Thus projection is not necessary for stability within this tested protocol; this is not a general training-stability conclusion.

D.10 PROJECTION-FREE CONTRACTION DIAL (FULL RESULTS)

The depth-mismatch summary in Appendix F.1 gives the high-level takeaway. Table 15 below provides the full numeric results at $8\times$ depth mismatch.

D.11 STABILITY–EXPRESSIVITY TRADEOFF (FULL RESULTS)

This appendix provides an exploratory stability–expressivity analysis. **Important:** Success rates here ($\approx 6.7\%$) are *not* comparable to Table 7 ($\approx 93\%$) because they use different evaluation protocols: Exp5 uses *matched compute* ($n_{\text{eval}}=2$, *proj. off*) to isolate contraction-related effects, while Table 7 uses $n_{\text{eval}}=2$ with mixed projection settings (no-contraction: $R=10$; contraction: *proj. off*). The low matched-compute success in Exp5 reflects the inherent difficulty of solving puzzles with only 2 unroll steps under uniform projection-off evaluation, not model deficiency.

Using inference-time scaling (scales $\in \{1.0, 0.85, 0.70, 0.55\}$, *proj. off*, value-head spectral normalization disabled), we observe that stability varies modestly across scale settings while task success

Table 14: Training-time projection ablation (4×4 trivial Sudoku, feasibility checker). Projection setting $\in \{\text{proj. off}, 10, 100\}$ crossed with contraction ON/OFF. All conditions pass the stated stability gate. Within this tested easy-suite protocol, success rates are comparable with *proj. off*; this is not a general training-stability claim.

Condition	Contraction	R	Success (%)	Stability Gate
No-Contr / $R=10$	OFF	10	94	PASS
No-Contr / $R=100$	OFF	100	96	PASS
No-Contr / <i>proj. off</i>	OFF	<i>proj. off</i>	96	PASS
Contr / $R=10$	ON	10	98	PASS
Contr / $R=100$	ON	100	96	PASS
Contr / <i>proj. off</i>	ON	<i>proj. off</i>	96	PASS

Table 15: Projection-free contraction dial results with *proj. off* and value-head spectral normalization disabled. Monotonic correlation with scale setting is statistically significant (Spearman ρ shown below).

Scale	$\hat{L}_{\text{preproj}}$	Argmax ($n_{\text{eval}}=8$) B0	Argmax ($n_{\text{eval}}=8$) B1	ΔV ($n_{\text{eval}}=8$) B0	Entropy
1.00	0.470 ± 0.326	0.887 ± 0.133	0.856 ± 0.156	0.267	2.61
0.85	0.383 ± 0.173	0.867 ± 0.148	0.862 ± 0.129	0.176	2.60
0.70	0.391 ± 0.084	0.880 ± 0.118	0.860 ± 0.102	0.532	2.65
0.55	0.490 ± 0.037	0.820 ± 0.080	0.793 ± 0.091	1.336	2.81
Spearman ρ (B0 Argmax, $n_{\text{eval}}=8$): $\rho = -0.866$ [95% CI: -1.000, -0.542]					
Spearman ρ (B1 Argmax, $n_{\text{eval}}=8$): $\rho = -0.923$ [95% CI: -1.000, -0.787]					

remains essentially flat ($\approx 6.7\%$); we do not observe a pronounced tradeoff in this regime. Table 16 provides the numeric results.

Figure 8 shows achieved $\hat{L}_z^{\text{pre}}$ vs. both stability and success.

E DECISION-RELEVANT HARD-SUITE RECORDS

This section retains the hard-suite task records and controlled projection $\times$ contraction aggregates used in the main narrative.

E.1 HARDER 4×4 SUDOKU (6–8 EMPTIES)

We extend evaluation to a harder suite with 6–8 empty cells, where the random-policy success rate drops to 0%.

E.2 DECONFOUNDED PROJECTION $\times$ CONTRACTION ON HARD 4×4

To isolate the mechanisms, we ran a controlled 2×2 factorial on the same no-mask hard 4×4 suite.

The projection contrast raises success from 36.2% to 49.2% (+13.0pp averaged over contraction status). It is a composite algorithmic effect spanning the forward pass and training dynamics (changes to f_θ , ε_0 , and the optimizer trajectory), which (70) does not model. Averaging over projection status, contraction raises success from 41.6% to 43.8% (+2.2pp). The source-reported $p \approx 0.6$ is not independently auditable because the retained package lacks the per-seed factorial artifact and test script; the aggregate contrast is inconclusive and does not support a variance-reduction claim.

F SECONDARY ROBUSTNESS DIAGNOSTICS

These depth-mismatch curves and discussion details are secondary diagnostics and do not establish hard-suite task performance.

Table 16: Exp5: Stability–Expressivity Analysis. Success remains essentially flat across scale settings ($\approx 6.7\%$); no clear tradeoff observed in this regime.

Scale	$\hat{L}_{\text{preproj}}$	Stability (argmax@ $8\times$)	Success (trivial)	$\Delta V@8\times$
1.00	0.801 ± 0.238	0.700 ± 0.085	0.067 ± 0.017	0.528
0.85	0.562 ± 0.136	0.730 ± 0.057	0.070 ± 0.014	0.439
0.70	0.489 ± 0.073	0.770 ± 0.098	0.067 ± 0.017	0.458
0.55	0.524 ± 0.031	0.730 ± 0.107	0.067 ± 0.012	0.680

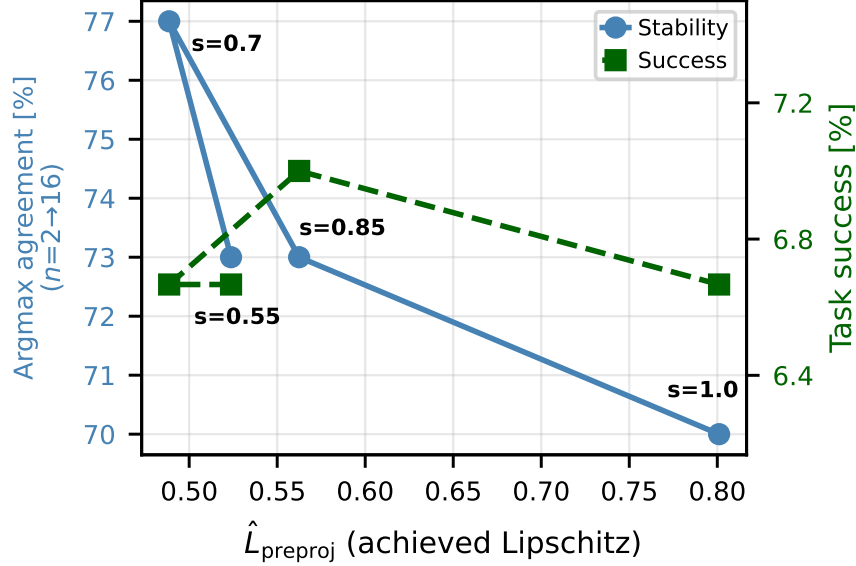


Figure 8: Achieved $\hat{L}_z^{\text{pre}}$ (x-axis) vs. argmax agreement [%] ($n=2 \rightarrow 16$, left y-axis, blue) and task success [%] (right y-axis, green). Scale annotations show s . **Note:** Success rates use matched compute ($n=2$, proj. off); Table 7 uses $n=2$ with mixed projection settings (see text).

F.1 DEPTH-MISMATCH ROBUSTNESS (ISOLATION STUDY)

To probe evaluation-time robustness as compute changes, we evaluate checkpoints trained at $n_{\text{train}}=2$ under deeper unrolls $n_{\text{eval}} \in \{4, 8, 16\}$ on the 1–4-empty suite and aggregate over seeds 41–50. On B0 initial states at fixed $4\times$ mismatch ($n=2 \rightarrow 8$), contraction lowers Δ_V from 0.166 to 0.064 and raises argmax agreement from 90.2% to 95.5%; Δ_π stays near zero. This is finite-batch diagnostic contact with the episodic setting, not theorem validation, and it does not speak to persistent- z behavior on the hard suite.

Figure 10 shows a clear robustness gain from contraction under deeper evaluation. The detailed tables (Appendix D.4) show that moving from *proj. off* to $R=10$ yields the larger reduction in Δ_V , while Appendix D.6 gives the same qualitative pattern on B1 successor states with higher noise.

F.2 CONTROLLED 2×2 (CONTRACTION $\times$ PROJECTION) ON HARD SUITE (FULL CURVES)

Historical hard-suite comparisons mixed projection settings and therefore confounded contraction with clipping. To deconfound, we ran a controlled 2×2 factorial (contraction ON/OFF $\times$ projection ON/OFF) on the same hard 4×4 Sudoku suite under the no-mask protocol (6–8 empties, 20k steps, 10 seeds). Figure 9 shows the full learning curves; Figure 11 summarizes final performance.

Three readings follow from the aggregate 10-seed table. Enabling projection raises mean success from 36.2% to 49.2% (+13.0pp averaged over contraction status), a composite algorithmic contrast.

2×2 Controlled: Contraction × Projection
(Hard 4×4 Sudoku, 6-8 empties, T=16)

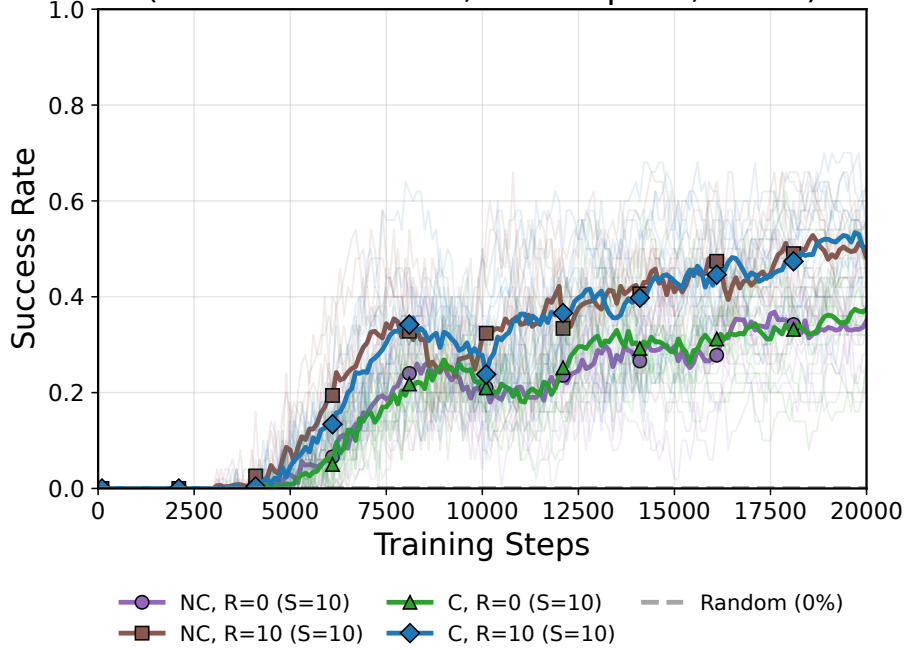


Figure 9: Controlled no-mask hard 4×4 2×2 learning curves on 6–8 empties: success rate (feasible within $T=16$) vs. training steps (20k steps, greedy evaluation, 10 seeds per cell). Final success rates are 35.0% (NC, proj. off), 48.2% (NC, $R=10$), 37.4% (C, proj. off), and 50.2% (C, $R=10$). Random remains at 0%. Projection shifts both contraction settings upward by about 13 percentage points, while contraction adds only a small average effect.

Table 17: Controlled no-mask hard 4×4 2×2 (10 seeds per cell). The projection contrast is larger; the contraction contrast is small and inconclusive.

Condition	Success (%)
No contraction, proj. off	35.0 ± 8.9
No contraction, $R=10$	48.2 ± 7.1
Contraction, proj. off	37.4 ± 7.8
Contraction, $R=10$	50.2 ± 10.5

Contraction adds only a small average effect: at *proj. off* it changes success from 35.0% to 37.4% (+2.4pp), and at $R=10$ from 48.2% to 50.2% (+2.0pp). The highest-mean cell is C+ $R=10$ (50.2% ± 10.5%) versus 48.2% ± 7.1% for no contraction at $R=10$; this factorial does not establish a contraction benefit or variance reduction.

The controlled table replaces the older asymmetric narrative: projection has the larger composite success contrast, while contraction has only a small, inconclusive average contrast. The separate depth-mismatch result concerns value stability, not hard-suite return.

F.3 DISCUSSION DETAILS

The stability experiments answer different questions, and the shortest consistent reading is: Appendix F.1 gives a binary contraction effect on *evaluation-time* depth-mismatch stability for B0 initial states; Appendix D.7 shows that active projection can dominate and mask that effect; Appendix D.9 shows projection is not required for *training* stability; and Appendix D.10 shows that once projection is disabled, the contraction proxy regains meaningful range. Appendix D.11 is exploratory only:

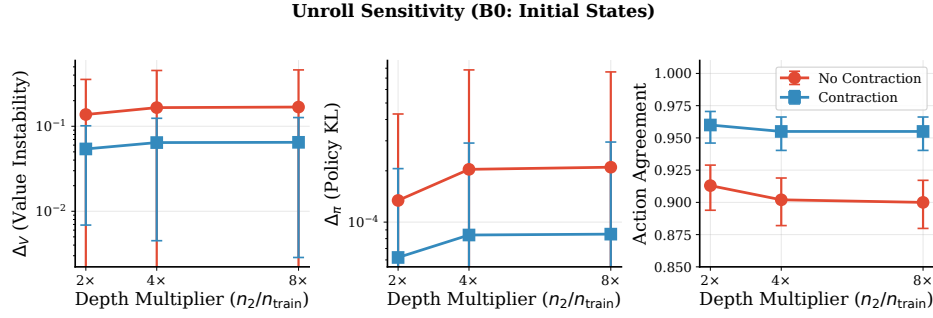


Figure 10: B0 depth-mismatch sensitivity for checkpoints trained at $n_{\text{train}}=2$ and evaluated at $n_{\text{eval}} \in \{4, 8, 16\}$. Both conditions use projection $R=10$ and 10 seeds. Contraction lowers Δ_V and improves argmax agreement; Δ_π stays near zero.

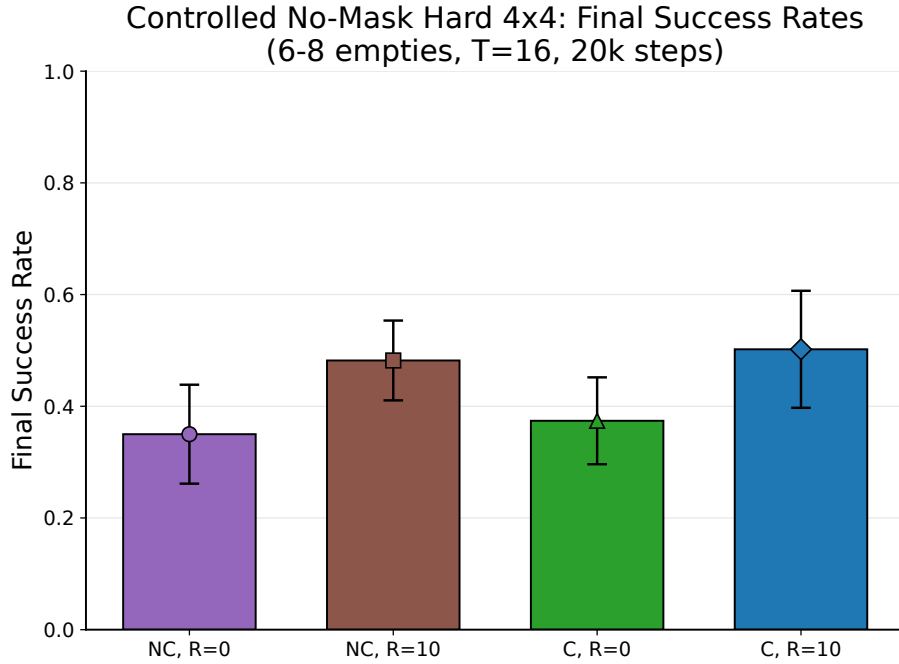


Figure 11: Controlled 2×2 (contraction $\times$ projection) on hard 4×4 Sudoku (6–8 empties, no-mask protocol, $T=16$, 20k steps, 10 seeds). Final success rates are **No Contraction + proj. off:** $35.0\% \pm 8.9\%$; **No Contraction + $R=10$:** $48.2\% \pm 7.1\%$; **Contraction + proj. off:** $37.4\% \pm 7.8\%$; **Contraction + $R=10$:** $50.2\% \pm 10.5\%$. The aggregate projection contrast is +13.0pp; contraction adds +2.2pp and does not reduce variance under projection.

its success rates ($\approx 6.7\%$) use matched compute ($n=2$, proj. off) and are not directly comparable to Table 7, which mixes projection settings.

These appendices show two observed contrasts: projection changes success and depth-stability diagnostics through coupled forward-pass and training channels, while disabling projection exposes a wider contraction-proxy range for inference-time checks. They do not isolate a projection training-stability effect or establish that contraction improves hard-suite performance.

A naive reading of the older asymmetric comparison would suggest that enforcing $L_z < 1$ is strongly harmful on harder tasks. The controlled no-mask 10-seed 2×2 does not support that reading. Projection changes mean success from 36.2% to 49.2%, while contraction changes it only from 41.6% to 43.8% averaged over projection settings (+2.2pp) and does not reduce variance under projection. Contraction is a sufficient error-control condition, but these data do not demonstrate it as a hard-task performance dial.

Table 18: Artifact status for the missing interaction-equalized comparison.

Run family	UPI-TRM arm	Standalone prove- nance	Claim status
Source baseline-only audit	Absent	Absent	Numerical rows ex- cluded
Required UPI-TRM vs. PPO audit	Required	Not run	Missing experiment

G PROTOCOL AND THEORY-ALIGNED RUN RECORDS

This section records the interaction-budget audit, headline significance data, and the theory-aligned episodic hard-suite run. The common-budget baseline table lacks standalone provenance in the retained package and is marked accordingly.

G.1 INTERACTION-EQUALIZED BASELINE AUDIT

Table 1 reports the nominal 20k outer-step protocol. Because PPO, A2C, and DQN collect 64, 32, and 4 transitions per outer step, it does not equalize interactions. The source manuscript recorded baseline-only common-budget rows, but their standalone logs are absent. We therefore remove those unauditable numbers from active claims (they remain recoverable in Git history) and retain only the artifact-status audit below.

There is therefore no numerical interaction-matched UPI-TRM versus PPO result in this paper.

G.2 HEADLINE SIGNIFICANCE-TEST DATA

The significance tests in Section 7.1 use the final per-seed success rates from the nominal 20k outer-step hard-suite protocol. For UPI-TRM, the 10 seed-level success rates are

0.56, 0.50, 0.68, 0.50, 0.64, 0.70, 0.74, 0.42, 0.38, 0.62,

with mean 57.4% and sample standard deviation 12.2%. For the architecture-matched TRM+PPO baseline, the corresponding rates are

0.30, 0.32, 0.12, 0.54, 0.12, 0.56, 0.20, 0.32, 0.42, 0.30,

with mean 32.0% and sample standard deviation 15.3%. Applying Welch’s t -test to these seed-level values gives $t \approx 4.10$, $df \approx 17.1$, two-sided $p \approx 7.3 \times 10^{-4}$, and a 95% confidence interval of [12.4, 38.5] percentage points for the mean difference. A 100,000-sample bootstrap over seeds gives a 95% interval of [13.8, 36.6] percentage points, and a permutation test gives $p \approx 1.2 \times 10^{-3}$.

G.3 THEORY-ALIGNED EPISODIC- z RERUN ON THE HARD SUITE

We ran a pre-registered, theory-aligned rerun on the same no-mask hard 4×4 Sudoku suite used in Table 1 (6–8 empties, $T=16$, 20k environment steps, seeds 41–50). The rerun starts from the persistent- z hard-suite configuration and jointly changes four factors: episodic- z latents, exact baseline summation, exact K -step targets, and explicit conservative mixtures. The architecture-matched evaluation interface and the inherited persistent- z hyperparameters ($\gamma=0.99$, $K=1$, $n=2$) are unchanged. Diagnostics are computed on a frozen 256-instance closure batch (seed 1729). The fixed test split of the canonical 6–8-empty dataset contains only 99 viable puzzles, so the closure batch is sampled from the `train/` split with an independent RNG path; this deviation affects only the diagnostic batch, not the task-evaluation pool or the training seeds.

The paper tables below curate the main quantities; full per-seed values for all nine protocol diagnostics are recorded in `EPISODIC_Z_RUN_LOG.md` and `results/episodic_z_hard_suite_20k_seed41_50/` (diagnostics_summary.json, cpi_penalty_grid.json, and per_seed_diagnostics.jsonl).

Table 19 reports the final task outcome and the main diagnostics at 20k. The rerun reaches 0.0% success over 10 seeds, so by the locked decision rule it falls into Branch D: $S_E = 0.0\% < 0.85 \times$

Table 19: Pre-registered theory-aligned episodic- z rerun at 20k on the no-mask hard 4×4 suite. Means $\pm$ sample std over 10 seeds unless noted.

Quantity	Value
Success (%)	0.0 ± 0.0
Return	-21.733 ± 0.512
Score	3.906 ± 0.606
Invalid	0.000 ± 0.000
Decision branch	D ($0.0\% < 0.85 \times 57.4\% = 48.79\%$)
$\varepsilon_{\text{cent}}$ @ 20k	$(1.782 \pm 0.301) \times 10^{-6}$
$\varepsilon_{\text{res},n}$ @ 20k	5.047 ± 0.175
projection-active rate @ 20k	1.0
ρ_R @ 20k	$45.2547 \pm 1.3 \times 10^{-5}$
finite-seed $\hat{L}_z^{\text{post}}$ @ 20k	0.600 ± 0.104 over 8/10 seeds
saturating seeds	41, 46 ($\hat{L}_z^{\text{post}} \geq 1$)

Table 20: Predicted CPI penalty at 20k on the episodic- z closure batch. Finite-subset mean $\pm$ sample std is computed over seeds 42, 43, 44, 45, 47, 48, 49, and 50. The all-seed mean is $+\infty$ at every α because seeds 41 and 46 have $\hat{L}_z^{\text{post}} \geq 1$.

α	finite-subset predicted penalty	all-seed mean
0.05	5215.11 ± 164.00	$+\infty$
0.10	10809.78 ± 303.85	$+\infty$
0.20	23137.80 ± 561.71	$+\infty$
0.40	52348.56 ± 1387.43	$+\infty$

57.4% = 48.79%. The persistent- z value $S_P = 57.4\%$ in Table 1 is measured under the nominal 20k outer-step protocol of Section 7.1, which uses a larger environment-interaction budget than this rerun; the Branch D margin holds under either budget framing. We therefore retain persistent- z as the headline empirical result and use the episodic- z rerun only for finite-batch diagnostics on the frozen closure batch.

At 20k, the finite-batch centering defect is numerically small, projection is active on every sampled closure-batch state, and the held-out finite-depth Bellman residual remains large. None of these finite-batch observations is a uniform certificate. Two seeds (41 and 46) have $\hat{L}_z^{\text{post}} \geq 1$, so the all-seed mean predicted CPI penalty is literally $+\infty$ at every $\alpha \in \{0.05, 0.10, 0.20, 0.40\}$. For interpretability, Table 20 also reports finite-seed summaries over the remaining 8/10 seeds.

H NOTATION SUMMARY

We summarize notation here for convenience. Table 21 lists additional symbols used in proofs.

I CONSTRUCTIVE DEFINITION OF THE ADVANTAGE-EVALUATION CLOSURE

Definition I.1 (Reachable set). For a policy π , let $\mathcal{R}_\pi$ be a measurable carrier of every state marginal: $(\rho P_\pi^t)(\mathcal{R}_\pi) = 1$ for all $t \geq 0$, where P_π is the policy-induced Markov kernel. In a finite or countable MDP we take the exact set of states reached with positive probability. In a general state space the carrier is understood up to sets that are null for all these marginals.

Definition I.2 (Advantage-evaluation closure). Fix a current policy π and a candidate policy π_{cand} . Let $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ be a measurable set containing $\mathcal{R}_\pi$ and s_{abs} such that

- for every $s \in \mathcal{R}_\pi$ and every action in $\text{supp}(\pi(\cdot|s)) \cup \text{supp}(\pi_{\text{cand}}(\cdot|s))$, the next state lies in $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ almost surely; and
- for every $s \in \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ and $a \in \text{supp}(\pi(\cdot|s))$, $P(\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} | s, a) = 1$.

Table 21: Additional symbols and notations

Symbol	Meaning
$z^{(j)}(s), U_j(s)$	Latent and scalar value after finite depth j
$U_*(s)$	Fixed-point value used only in the contractive specialization
$B_{n,m}$	Finite path/discrepancy plus depth- m residual certificate
$\tilde{U}_n(\bar{s}), \tilde{U}_*(\bar{s})$	Persistent finite-depth value / memoryless reference
$\varepsilon_{\text{res},n}$	Episodic finite-depth Bellman residual
$\bar{\varepsilon}_{\text{res},n}$	Persistent augmented-state finite-depth residual
$\bar{\varepsilon}_{\text{res},*}$	Augmented residual of the fixed-point reference
$L_{z^*}, \Delta y_{\text{max}}, E_0$	Slow-drift parameters
$\varepsilon_A, \bar{\varepsilon}_A$	Episodic / augmented advantage error bounds
$\varepsilon_{\text{res}}^*$	Architectural residual at U_*
ε_{CPI}	CPI constant
$\kappa_n = L_z^n$	Contraction after n steps
$\mathcal{R}_\Pi$	Union of reachable sets for policy family Π
$\ f\ _{\infty, \Pi}$	Sup-norm over $\mathcal{R}_\Pi$
$B_V(s)$	Baseline: $\mathbb{E}_{b \sim \pi}[\hat{Q}_V(s, b)]$

We write $\|f\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$ for the essential supremum of $|f|$ on this set (including the absorbing boundary). The essential-supremum reference measure must dominate the transition laws appearing above; in the finite experiments this is simply counting measure and the norm is an ordinary maximum.

For completeness, in a finite or countable MDP the following construction gives the exact closure in Definition I.2. For general kernels, the same recursion uses measurable carriers of the indicated transition laws rather than point masses; the theorem requires an admissible measurable closure, not a pointwise “positive density” test or a unique minimal carrier.

Fix a current policy π and a candidate policy π_{cand} . Define the *one-step deviation* set

$$\mathcal{R}_{\pi, \pi_{\text{cand}}}^+ := \mathcal{R}_\pi \cup \{s' : \exists s \in \mathcal{R}_\pi, \exists a \in \text{supp}(\pi(\cdot|s)) \cup \text{supp}(\pi_{\text{cand}}(\cdot|s)) \text{ with } P(s' | s, a) > 0\}. \quad (47)$$

Let $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ denote the smallest superset of $\mathcal{R}_{\pi, \pi_{\text{cand}}}^+$ that is closed under forward dynamics of π :

$$\begin{aligned} \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} &:= \bigcup_{t \geq 0} \mathcal{R}^{(t)}, \quad \mathcal{R}^{(0)} = \mathcal{R}_{\pi, \pi_{\text{cand}}}^+, \\ \mathcal{R}^{(t+1)} &= \mathcal{R}^{(t)} \cup \{s' : \exists s \in \mathcal{R}^{(t)}, \\ &\quad a \in \text{supp}(\pi(\cdot|s)), P(s' | s, a) > 0\}. \end{aligned} \quad (48)$$

The sup-norm $\|f\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$ used throughout is the one introduced in Definition I.2.

J SHAPING CONVENTION AND ABSORBING-STATE NORMALIZATION

We use potential-based shaping with potential $\Phi(s) = c(x, y)$ for $s = (x, y)$ and $\Phi(s_{\text{abs}}) = C_{\text{max}}$, and define

$$r(s, a, s') = r_0(s, a, s') + \gamma \Phi(s') - \Phi(s), \quad (49)$$

applying shaping on *all* transitions, including the absorbing self-loop.

For the absorbing self-loop, taking $r_0(s_{\text{abs}}, a, s_{\text{abs}}) = 0$ gives

$$r(s_{\text{abs}}, a, s_{\text{abs}}) = \gamma \Phi(s_{\text{abs}}) - \Phi(s_{\text{abs}}) = (\gamma - 1)C_{\text{max}}.$$

Thus every policy satisfies the fixed shaped boundary value

$$V^\pi(s_{\text{abs}}) = \sum_{t=0}^{\infty} \gamma^t (\gamma - 1) C_{\text{max}} = -C_{\text{max}}.$$

If one prefers the standard normalization $V(s_{\text{abs}}) = 0$, use the shifted potential $\tilde{\Phi}(s) := \Phi(s) - C_{\max}$. This adds the constant reward $(1 - \gamma)C_{\max}$ to every transition and shifts all returns by $+C_{\max}$ independently of the policy. Accordingly, performance differences and optimal policies are unchanged.

K CPI DERIVATION FOR MIXTURE POLICIES

We derive the CPI bound (29) specialized to mixture policies $\pi_{\text{new}} = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$, following the framework of Kakade & Langford (2002).

Setup. Define the *advantage function under the greedy policy* π_{cand} :

$$\delta(s) := \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot | s)}[A^\pi(s, a)]. \quad (50)$$

Note that $|\delta(s)| \leq \varepsilon_{\text{CPI}} := \max_s |\delta(s)|$ by definition.

For the mixture policy $\pi_{\text{new}} = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$:

$$\mathbb{E}_{a \sim \pi_{\text{new}}(\cdot | s)}[A^\pi(s, a)] = (1 - \alpha) \underbrace{\mathbb{E}_{a \sim \pi}[A^\pi(s, a)]}_{=0} + \alpha \mathbb{E}_{a \sim \pi_{\text{cand}}}[A^\pi(s, a)] \quad (51)$$

$$= \alpha \delta(s). \quad (52)$$

Action-set convention. If the available edit actions depend on the plan (and hence on the state $s = (x, y)$), we interpret $\pi(\cdot | s)$ and $\pi_{\text{cand}}(\cdot | s)$ as distributions on the same per-state action set $\mathcal{A}(s)$. Equivalently, we may embed both policies into a fixed superset action space by assigning probability 0 to invalid actions. Under either convention, $\|\pi'(\cdot | s) - \pi(\cdot | s)\|_1 \leq 2$ holds for all s . **Performance difference lemma.** The standard performance difference lemma (Kakade & Langford, 2002) gives:

$$\eta(\pi_{\text{new}}) - \eta(\pi) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi_{\text{new}}}} [\mathbb{E}_{a \sim \pi_{\text{new}}}[A^\pi(s, a)]] = \frac{\alpha}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi_{\text{new}}}} [\delta(s)]. \quad (53)$$

Distribution shift bound. We need to relate $\mathbb{E}_{s \sim d_{\pi_{\text{new}}}} [\delta(s)]$ to $\mathbb{E}_{s \sim d_\pi} [\delta(s)]$. The key is to bound $\|d_{\pi_{\text{new}}} - d_\pi\|_1$.

Lemma K.1 (Occupancy measure coupling). *For any two policies π' and π in an MDP with discount γ :*

$$\|d_{\pi'} - d_\pi\|_1 \leq \frac{\gamma}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi'}} [\|\pi'(\cdot | s) - \pi(\cdot | s)\|_1]. \quad (54)$$

Proof. Recall the discounted occupancy satisfies the fixed-point equation

$$d_\pi = (1 - \gamma)\rho + \gamma d_\pi P_\pi, \quad d_{\pi'} = (1 - \gamma)\rho + \gamma d_{\pi'} P_{\pi'}.$$

Subtracting gives

$$d_{\pi'} - d_\pi = \gamma(d_{\pi'} P_{\pi'} - d_\pi P_\pi) = \gamma(d_{\pi'}(P_{\pi'} - P_\pi) + (d_{\pi'} - d_\pi)P_\pi).$$

Taking ℓ_1 norms and using that P_π is a Markov operator (so it is non-expansive in ℓ_1),

$$\|d_{\pi'} - d_\pi\|_1 \leq \gamma \|d_{\pi'}(P_{\pi'} - P_\pi)\|_1 + \gamma \|(d_{\pi'} - d_\pi)P_\pi\|_1 \leq \gamma \|d_{\pi'}(P_{\pi'} - P_\pi)\|_1 + \gamma \|d_{\pi'} - d_\pi\|_1.$$

Rearranging yields

$$\|d_{\pi'} - d_\pi\|_1 \leq \frac{\gamma}{1 - \gamma} \|d_{\pi'}(P_{\pi'} - P_\pi)\|_1.$$

Finally,

$$\begin{aligned}
\|d_{\pi'}(P_{\pi'} - P_{\pi})\|_1 &= \left\| \sum_s d_{\pi'}(s) (P_{\pi'}(\cdot|s) - P_{\pi}(\cdot|s)) \right\|_1 \\
&\leq \sum_s d_{\pi'}(s) \|P_{\pi'}(\cdot|s) - P_{\pi}(\cdot|s)\|_1 \\
&= \mathbb{E}_{s \sim d_{\pi'}} [\|P_{\pi'}(\cdot|s) - P_{\pi}(\cdot|s)\|_1] \\
&= \mathbb{E}_{s \sim d_{\pi'}} \left[\left\| \sum_a (\pi'(a|s) - \pi(a|s)) P(\cdot|s, a) \right\|_1 \right] \\
&\leq \mathbb{E}_{s \sim d_{\pi'}} \left[\sum_a |\pi'(a|s) - \pi(a|s)| \underbrace{\|P(\cdot|s, a)\|_1}_{=1} \right] \\
&= \mathbb{E}_{s \sim d_{\pi'}} [\|\pi'(\cdot|s) - \pi(\cdot|s)\|_1], \tag{55}
\end{aligned}$$

which gives the claim. $\square$

For our mixture:

$$\|\pi_{\text{new}}(\cdot|s) - \pi(\cdot|s)\|_1 = \alpha \|\pi_{\text{cand}}(\cdot|s) - \pi(\cdot|s)\|_1 \leq 2\alpha, \tag{56}$$

since $\|p - q\|_1 \leq 2$ for any two probability distributions p, q on the same action space.

Thus:

$$\|d_{\pi_{\text{new}}} - d_{\pi}\|_1 \leq \frac{2\gamma\alpha}{1-\gamma}. \tag{57}$$

Combining the bounds. Write:

$$\mathbb{E}_{s \sim d_{\pi_{\text{new}}}} [\delta(s)] = \mathbb{E}_{s \sim d_{\pi}} [\delta(s)] + (\mathbb{E}_{s \sim d_{\pi_{\text{new}}}} [\delta(s)] - \mathbb{E}_{s \sim d_{\pi}} [\delta(s)]). \tag{58}$$

The second term is bounded by:

$$|\mathbb{E}_{s \sim d_{\pi_{\text{new}}}} [\delta(s)] - \mathbb{E}_{s \sim d_{\pi}} [\delta(s)]| \leq \|d_{\pi_{\text{new}}} - d_{\pi}\|_1 \cdot \sup_s |\delta(s)| \leq \frac{2\gamma\alpha}{1-\gamma} \varepsilon_{\text{CPI}}. \tag{59}$$

Therefore:

$$\eta(\pi_{\text{new}}) = \eta(\pi) + \frac{\alpha}{1-\gamma} \mathbb{E}_{s \sim d_{\pi_{\text{new}}}} [\delta(s)] \tag{60}$$

$$\geq \eta(\pi) + \frac{\alpha}{1-\gamma} \mathbb{E}_{s \sim d_{\pi}} [\delta(s)] - \frac{\alpha}{1-\gamma} \cdot \frac{2\gamma\alpha}{1-\gamma} \varepsilon_{\text{CPI}} \tag{61}$$

$$= L_{\pi}(\pi_{\text{new}}) - \frac{2\gamma\varepsilon_{\text{CPI}}}{(1-\gamma)^2} \alpha^2, \tag{62}$$

where in the last step we used $L_{\pi}(\pi_{\text{new}}) = \eta(\pi) + \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_{\pi}, a \sim \pi_{\text{new}}} [A^{\pi}(s, a)] = \eta(\pi) + \frac{\alpha}{1-\gamma} \mathbb{E}_{s \sim d_{\pi}} [\delta(s)]$.

This establishes (29).

Corollary K.2 (CPI with a centering defect). *Use the setup of Theorem 6.5, but replace the exact-centering condition (18) by*

$$\varepsilon_{\text{cent}} := \sup_{s \in \mathcal{R}_{\pi}} \left| \mathbb{E}_{a \sim \pi(\cdot|s)} [\hat{A}(s, a)] \right| < \infty.$$

Then

$$\eta(\pi_{\text{new}}) \geq \hat{L}_{\pi}(\pi_{\text{new}}) - \frac{(1-\alpha)\varepsilon_{\text{cent}} + \alpha\varepsilon_{A, \text{cand}}}{1-\gamma} - \frac{2\varepsilon_{\text{CPI}}\gamma}{(1-\gamma)^2} \alpha^2. \tag{63}$$

Proof. The CPI shell gives

$$\eta(\pi_{\text{new}}) \geq L_{\pi}(\pi_{\text{new}}) - \frac{2\varepsilon_{\text{CPI}}\gamma}{(1-\gamma)^2} \alpha^2.$$

For each $s \in \mathcal{R}_\pi$,

$$\begin{aligned} & \mathbb{E}_{a \sim \pi_{\text{new}}(\cdot|s)}[A^\pi(s, a) - \hat{A}(s, a)] \\ &= (1 - \alpha) \mathbb{E}_{a \sim \pi(\cdot|s)}[A^\pi(s, a) - \hat{A}(s, a)] + \alpha \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)}[A^\pi(s, a) - \hat{A}(s, a)]. \end{aligned}$$

Since $\mathbb{E}_{a \sim \pi}[A^\pi(s, a)] = 0$, the first term is bounded below by $-(1 - \alpha)\varepsilon_{\text{cent}}$. By the definition of $\varepsilon_{A, \text{cand}}$, the second term is bounded below by $-\alpha\varepsilon_{A, \text{cand}}$. Therefore

$$L_\pi(\pi_{\text{new}}) \geq \hat{L}_\pi(\pi_{\text{new}}) - \frac{(1 - \alpha)\varepsilon_{\text{cent}} + \alpha\varepsilon_{A, \text{cand}}}{1 - \gamma}.$$

Combining the two displays proves (63). $\square$

K.1 PROOF OF COROLLARY 6.2

Proof. Because $\mathcal{T}_K^\pi$ is a γ^K -contraction with fixed point V^π ,

$$\begin{aligned} \|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} &\leq \|U_n - \mathcal{T}_K^\pi U_n\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + \|\mathcal{T}_K^\pi U_n - \mathcal{T}_K^\pi V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \\ &\leq \varepsilon_{\text{res}, n} + \gamma^K \|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \end{aligned}$$

Rearranging gives (10). Lemma L.2 applied to the exact one-step centered estimator $\hat{A}_{U_n}$ then yields (11); by definition, $\varepsilon_{A, \text{cand}} \leq \varepsilon_A$. Substituting that bound into Theorem 6.5 gives (12). $\square$

Remark K.3 (Residual bridge via proxy quantities). The held-out U_n Bellman residual is a practical proxy for $\varepsilon_{\text{res}}^*$, but the global sup-norm quantity below is not directly measurable from finite data; it upper-bounds $\varepsilon_{\text{res}}^*$ up to the same truncation gap:

$$\varepsilon_{\text{res}}^* \leq \|U_n - \mathcal{T}_K^\pi U_n\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + (1 + \gamma^K) \|U_n - U_*\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \quad (64)$$

This follows by adding and subtracting U_n inside $U_* - \mathcal{T}_K^\pi U_*$ and using K -step Bellman contraction on the final term. Combining (64) with Proposition A.2 gives

$$\varepsilon_{\text{res}}^* \leq \|U_n - \mathcal{T}_K^\pi U_n\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + (1 + \gamma^K) L_V \frac{L_z^n}{1 - L_z} C_z. \quad (65)$$

L ADVANTAGE ESTIMATION AND BOUNDS

$$\hat{Q}_V(s, a) = \mathbb{E}_{s'}[r(s, a, s') + \gamma V(s')], \quad (66)$$

$$\hat{A}_V(s, a) = \hat{Q}_V(s, a) - \mathbb{E}_{b \sim \pi(\cdot|s)}[\hat{Q}_V(s, b)]. \quad (67)$$

Remark L.1 (Centering and baseline computation). The linear-in- α $\varepsilon_{A, \text{cand}}$ term in Theorem 6.5 relies on the *statewise centering* property

$$\mathbb{E}_{a \sim \pi(\cdot|s)}[\hat{A}(s, a)] = 0 \quad \text{for all } s.$$

This holds identically when $\hat{A}$ is defined as in (67) with the baseline computed exactly as $B_V(s) = \mathbb{E}_{b \sim \pi(\cdot|s)}[\hat{Q}_V(s, b)]$ (e.g., by summation over a finite action space).

If the baseline is approximated, centering may fail. We quantify this by the *centering defect*

$$\varepsilon_{\text{cent}} := \sup_{s \in \mathcal{R}_\pi} \left| \mathbb{E}_{a \sim \pi(\cdot|s)}[\hat{A}(s, a)] \right|.$$

Theorem 6.5 corresponds to the ideal case $\varepsilon_{\text{cent}} = 0$.

Lemma L.2 (Value to one-step advantage error). *Let $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ be the advantage-evaluation closure from Definition I.2, and let $\delta_V := \|V - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$. Then for all (s, a) with $s \in \mathcal{R}_\pi$ and $a \in \text{supp}(\pi(\cdot|s)) \cup \text{supp}(\pi_{\text{cand}}(\cdot|s))$,*

$$|\hat{A}_V(s, a) - A^\pi(s, a)| \leq 2\gamma \delta_V.$$

The factor 2γ is tight worst-case (error appears in both Q and baseline). In practice, errors in $\widehat{Q}_V(s, a)$ and the baseline $\mathbb{E}_\pi[\widehat{Q}_V]$ often have the same sign and partially cancel.

Remark L.3 (Extensions to K -step and GAE-style advantages). Lemma L.2 is stated for the *one-step* estimator (67), hence the factor 2γ . If instead one defines a K -step action-value estimate

$$\widehat{Q}_V^{(K)}(s, a) := \mathbb{E} \left[\sum_{t=0}^{K-1} \gamma^t r_t + \gamma^K V(s_K) \mid s_0 = s, a_0 = a, \pi \right],$$

and the corresponding centered advantage $\widehat{A}_V^{(K)}(s, a) := \widehat{Q}_V^{(K)}(s, a) - \mathbb{E}_{b \sim \pi(\cdot|s)}[\widehat{Q}_V^{(K)}(s, b)]$ (with the baseline computed exactly), then the same argument yields

$$|\widehat{A}_V^{(K)}(s, a) - A^\pi(s, a)| \leq 2\gamma^K \delta_V.$$

In practice we may use λ -return / GAE-style estimators (weighted mixtures of multi-step returns), which admit analogous bounds with a coefficient depending on (γ, λ) ; throughout, we absorb such estimator-dependent constants into the uniform advantage error parameter ε_A in Assumption 6.4.

Corollary L.4 (Advantage error in terms of theorem parameters). *Under Theorem 6.1 and the exact-centering conditions of Lemma L.2,*

$$\varepsilon_A \leq 2\gamma B_{n,m}. \quad (68)$$

Under the additional contractive and Lipschitz assumptions, taking $m \rightarrow \infty$ yields

$$\varepsilon_A \leq \underbrace{\frac{2\gamma \varepsilon_{\text{res}}^*}{1 - \gamma^K}}_{\text{architectural}} + \underbrace{2\gamma L_V \frac{L_z^n}{1 - L_z}}_{\text{unrolling}} C_z. \quad (69)$$

Here ε_A bounds the advantage error on $\mathcal{R}_\pi$ for actions $a \in \text{supp}(\pi(\cdot|s)) \cup \text{supp}(\pi_{\text{cand}}(\cdot|s))$, and the intermediate value-error quantity in Lemma L.2 is taken over the advantage-evaluation closure $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ (Definition I.2).

M PROOFS

M.1 LEMMA M.1 (INTERIOR PROJECTION FIXED POINTS)

Lemma M.1 (Interior projection fixed points). *Let $\Pi_R : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be the Euclidean projection onto the closed ball $\{z : \|z\|_2 \leq R\}$ as in (23). Suppose $z^* = \Pi_R(u)$ for some $u \in \mathbb{R}^d$. If $\|z^*\|_2 < R$, then $z^* = u$ (i.e., the projection is inactive at u). Consequently, if z^* is a fixed point of $f_\theta(z, y, x) = \Pi_R(f_\theta(z, y, x))$ and $\|z^*\|_2 < R$, then $f_\theta(z^*, y, x) = z^*$ and Π_R is inactive at $f_\theta(z^*, y, x)$.*

Proof. By (23), if $\|u\|_2 \leq R$ then $\Pi_R(u) = u$. Since $z^* = \Pi_R(u)$ lies in the open ball $\|z^*\|_2 < R$, we have $z^* \neq R \frac{u}{\|u\|_2}$, so the second case of (23) cannot apply; hence $\|u\|_2 \leq R$ and $\Pi_R(u) = u = z^*$.

For the consequence: if $z^* = \tilde{f}_\theta(z^*, y, x) = \Pi_R(f_\theta(z^*, y, x))$ and $\|z^*\|_2 < R$, applying the first statement with $u = f_\theta(z^*, y, x)$ gives $f_\theta(z^*, y, x) = z^*$. $\square$

M.2 PROOF OF PROPOSITION A.2

Proof. Fix (x, y) . By Assumption 5.1, $T_{x,y}(z) = \tilde{f}_\theta(z, y, x)$ maps $\mathcal{Z}_{\text{inv}}$ to itself (forward-invariance) and is a contraction with modulus $L_z < 1$. Since $\mathcal{Z}_{\text{inv}}$ is a closed subset of the complete metric space $(\mathcal{Z}, \|\cdot\|) = (\mathbb{R}^d, \|\cdot\|_2)$, it is itself complete.

Banach's theorem yields unique fixed point $z^*(x, y) \in \mathcal{Z}_{\text{inv}}$ and linear convergence (26).

For the finite-unrolling bound with $z^{(0)} = z_{\text{init}}(x, y)$:

$$z^* - z^{(n)} = \sum_{k=n}^{\infty} (z^{(k+1)} - z^{(k)}).$$

Contraction gives $\|z^{(k+1)} - z^{(k)}\| \leq L_z^k \|z^{(1)} - z^{(0)}\|$, so:

$$\|z^* - z^{(n)}\| \leq \|z^{(1)} - z^{(0)}\| \sum_{k=n}^{\infty} L_z^k = \frac{L_z^n}{1 - L_z} \|z^{(1)} - z^{(0)}\|.$$

By definition (25) (using the projected update $\tilde{f}_\theta = \Pi_R \circ f_\theta$), we have $\|z^{(1)} - z^{(0)}\| = \|\tilde{f}_\theta(z_{\text{init}}, y, x) - z_{\text{init}}\| \leq C_z$. $\square$

M.3 PROOF OF LEMMA B.7

Proof. For $T_{x,y_t}^{\circ n}$ a κ_n -contraction (where $\kappa_n = L_z^n$):

$$\|z_{t+1} - z^*(x, y_t)\| = \|T_{x,y_t}^{\circ n}(z_t) - T_{x,y_t}^{\circ n}(z^*(x, y_t))\| \leq \kappa_n e_t.$$

By Assumption B.3:

$$\|z^*(x, y_t) - z^*(x, y_{t+1})\| \leq L_{z^*} \|y_{t+1} - y_t\|_{\mathcal{E}} \leq L_{z^*} \Delta y_{\max}.$$

Thus:

$$e_{t+1} = \|z_{t+1} - z^*(x, y_{t+1})\| \leq \|z_{t+1} - z^*(x, y_t)\| + \|z^*(x, y_t) - z^*(x, y_{t+1})\| \leq \kappa_n e_t + L_{z^*} \Delta y_{\max}.$$

Unrolling this recurrence:

$$e_t \leq \kappa_n^t e_0 + L_{z^*} \Delta y_{\max} \sum_{j=0}^{t-1} \kappa_n^j = \kappa_n^t e_0 + \frac{L_{z^*} \Delta y_{\max} (1 - \kappa_n^t)}{1 - \kappa_n}.$$

Using $e_0 \leq E_0$ and noting that this expression is maximized either at $t = 0$ (giving E_0) or as $t \rightarrow \infty$ (giving $\frac{L_{z^*} \Delta y_{\max}}{1 - \kappa_n}$), we have:

$$\sup_t e_t \leq \max \left(E_0, \frac{L_{z^*} \Delta y_{\max}}{1 - \kappa_n} \right) \leq E_0 + \frac{L_{z^*} \Delta y_{\max}}{1 - \kappa_n}.$$

For the used latent:

$$\|z_t^{(n)} - z^*(x, y_t)\| = \|T_{x,y_t}^{\circ n}(z_t) - z^*(x, y_t)\| = \|T_{x,y_t}^{\circ n}(z_t) - T_{x,y_t}^{\circ n}(z^*(x, y_t))\| \leq \kappa_n e_t,$$

where the second equality uses that $z^*(x, y_t)$ is the fixed point of T_{x,y_t} , hence also of $T_{x,y_t}^{\circ n}$, and the inequality follows from κ_n -contractivity of the n -fold composition. Thus:

$$\sup_t \|z_t^{(n)} - z^*(x, y_t)\| \leq \kappa_n \sup_t e_t \leq \kappa_n \max \left(E_0, \frac{L_{z^*} \Delta y_{\max}}{1 - \kappa_n} \right) = C_{\text{drift}}(n),$$

and the final “ $\leq$ ” in (42) gives the simpler additive upper bound. $\square$

M.4 PROOF OF THEOREM 6.1

Proof. Write $\|\cdot\|_{\infty}$ for the essential sup-norm on $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} \cup \{s_{\text{abs}}\}$. The measurability, boundedness, and almost-sure forward-invariance assumptions ensure that $\mathcal{T}_K^{\pi}$ maps the relevant bounded functions back to this space and is a γ^K -contraction in this norm. Since $V^{\pi} = \mathcal{T}_K^{\pi} V^{\pi}$,

$$\begin{aligned} \|U_m - V^{\pi}\|_{\infty} &\leq \|U_m - \mathcal{T}_K^{\pi} U_m\|_{\infty} + \|\mathcal{T}_K^{\pi} U_m - \mathcal{T}_K^{\pi} V^{\pi}\|_{\infty} \\ &\leq \|U_m - \mathcal{T}_K^{\pi} U_m\|_{\infty} + \gamma^K \|U_m - V^{\pi}\|_{\infty}. \end{aligned}$$

Rearranging and then applying the triangle inequality to $U_n - U_m + U_m - V^{\pi}$ proves (2).

For every nonabsorbing s , Lipschitzness of the scalar head and the telescoping latent path give

$$\begin{aligned} |U_n(s) - U_m(s)| &\leq L_V \|z^{(n)}(s) - z^{(m)}(s)\| \\ &\leq L_V \sum_{j=n}^{m-1} \|z^{(j+1)}(s) - z^{(j)}(s)\|. \end{aligned}$$

The left side is zero at s_{abs} because its value is depth independent. Taking the essential supremum and using $\text{ess sup}_s \sum_j g_j(s) \leq \sum_j \text{ess sup}_s g_j(s)$ proves (3). The latent norm is the same norm fixed in the theorem, so each summand has latent units and multiplication by L_V gives value units.

Under Assumption 5.1,

$$\|z^{(j+1)}(s) - z^{(j)}(s)\| \leq L_z^j \|z^{(1)}(s) - z^{(0)}(s)\| \leq L_z^j C_z.$$

Summing from $j = n$ through $m - 1$ yields (5). Proposition A.2 and the Lipschitz head give $\|U_m - U_*\|_\infty \rightarrow 0$. Moreover,

$$\|U_m - \mathcal{T}_K^\pi U_m\|_\infty - \|U_* - \mathcal{T}_K^\pi U_*\|_\infty \leq (1 + \gamma^K) \|U_m - U_*\|_\infty \rightarrow 0.$$

Taking $m \rightarrow \infty$ therefore proves (6). $\square$

Remark M.2 (Finite- R specialization of the truncation constant). Under the Appendix N initialization condition $\|z_{\text{init}}(x, y)\|_2 \leq R$ with $\mathcal{Z}_{\text{inv}} = \{z : \|z\|_2 \leq R\}$, the projection Π_R ensures $\|\tilde{f}_\theta(z_{\text{init}}(x, y), y, x)\|_2 \leq R$ for all (x, y) . Therefore

$$C_z = \sup_{(x, y)} \|\tilde{f}_\theta(z_{\text{init}}(x, y), y, x) - z_{\text{init}}(x, y)\|_2 \leq 2R.$$

The contractive specialization of Theorem 6.1 therefore gives

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{\varepsilon_{\text{res}}^*}{1 - \gamma^K} + 2L_V \frac{L_z^n}{1 - L_z} R.$$

This makes R explicit only through the finite-unrolling term; it does not identify the dependence of $\varepsilon_{\text{res}}^*$ on R .

M.5 PROJECTION-ACTIVE THEORY

Projection-aware CPI penalty (full statement). In the projection-active regime, assuming $L_z^{\text{sat}}(R) < 1$ in Proposition M.3, combining (7), (8), and $C_z \leq 2R$ yields a projection-aware version of (22):

$$\begin{aligned} \eta(\pi_{\text{new}}) &\geq \hat{L}_\pi(\pi_{\text{new}}) - \frac{2\alpha\gamma}{1 - \gamma} \left(\frac{\varepsilon_0 + (1 + \gamma^K)L_V(R)R}{1 - \gamma^K} + 2L_V(R) \frac{(L_z^{\text{sat}}(R))^n}{1 - L_z^{\text{sat}}(R)} R \right) \\ &\quad - \frac{2\varepsilon_{\text{CPI}}\gamma}{(1 - \gamma)^2} \alpha^2. \end{aligned} \tag{70}$$

Projection affects both the architectural and truncation terms; the finite- R sweep (Appendix D.5) shows this is a regime diagnostic, not a monotone tuning rule for R .

Proposition M.3 (Projection-induced contraction on a saturated annulus). *Let $B_R := \{z : \|z\|_2 \leq R\}$ and define the pre-projection local Lipschitz constant*

$$L_z^{\text{pre}}(R) := \sup_{(x, y)} \sup_{\substack{z, z' \in B_R \\ z \neq z'}} \frac{\|f_\theta(z, y, x) - f_\theta(z', y, x)\|_2}{\|z - z'\|_2}.$$

Assume there exists $\rho_R > R$ such that

$$\|f_\theta(z, y, x)\|_2 \geq \rho_R \quad \text{for all relevant } (x, y) \text{ and } z \in B_R.$$

Then the projected map $\tilde{f}_\theta = \Pi_R \circ f_\theta$ is Lipschitz on B_R with modulus

$$L_z^{\text{sat}}(R) \leq \frac{R}{\rho_R} L_z^{\text{pre}}(R).$$

In particular, if $(R/\rho_R) L_z^{\text{pre}}(R) < 1$, then the contraction part of Assumption 5.1 holds on B_R with $L_z \leq L_z^{\text{sat}}(R)$; if moreover $z_{\text{init}}(x, y) \in B_R$ for all (x, y) , then Assumption 5.1 holds in full on B_R .

Proof. For any u, v with $\|u\|_2, \|v\|_2 \geq \rho_R > R$,

$$\|\Pi_R(u) - \Pi_R(v)\|_2 = R \left\| \frac{u}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2.$$

Using

$$\left\| \frac{u}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2^2 = \frac{\|u - v\|_2^2 - (\|u\|_2 - \|v\|_2)^2}{\|u\|_2 \|v\|_2} \leq \frac{\|u - v\|_2^2}{\rho_R^2},$$

we obtain

$$\|\Pi_R(u) - \Pi_R(v)\|_2 \leq \frac{R}{\rho_R} \|u - v\|_2.$$

Applying this with $u = f_\theta(z, y, x)$ and $v = f_\theta(z', y, x)$ for $z, z' \in B_R$ gives

$$\|\tilde{f}_\theta(z, y, x) - \tilde{f}_\theta(z', y, x)\|_2 \leq \frac{R}{\rho_R} L_z^{\text{pre}}(R) \|z - z'\|_2. \quad \square$$

Proposition M.4 (Projection-aware architectural residual via a plan-independent latent anchor). *Fix any anchor map $z_{\text{anc}} : \mathcal{X} \rightarrow B_R$ that depends only on the instance x , and define*

$$B_{\text{anc}}(x, y) := V_\psi(z_{\text{anc}}(x), x), \quad \varepsilon_{\text{anc}} := \|B_{\text{anc}} - \mathcal{T}_K^\pi B_{\text{anc}}\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}.$$

Let

$$D_{\text{anc}}(R) := \sup_{(x, y)} \|z^*(x, y) - z_{\text{anc}}(x)\|_2, \quad L_V(R) := \sup_x \sup_{\substack{z, z' \in B_R \\ z \neq z'}} \frac{|V_\psi(z, x) - V_\psi(z', x)|}{\|z - z'\|_2}.$$

Then

$$\varepsilon_{\text{res}}^* \leq \varepsilon_{\text{anc}} + (1 + \gamma^K) L_V(R) D_{\text{anc}}(R). \quad (71)$$

In the special case $z_{\text{anc}}(x) \equiv 0$,

$$\varepsilon_{\text{res}}^* \leq \varepsilon_0 + (1 + \gamma^K) L_V(R) R, \quad \varepsilon_0 := \|V_\psi(0, x) - \mathcal{T}_K^\pi V_\psi(0, x)\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}.$$

Proof. Add and subtract B_{anc} :

$$U_* - \mathcal{T}_K^\pi U_* = (B_{\text{anc}} - \mathcal{T}_K^\pi B_{\text{anc}}) + (U_* - B_{\text{anc}}) + (\mathcal{T}_K^\pi B_{\text{anc}} - \mathcal{T}_K^\pi U_*).$$

Taking $\|\cdot\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$ and using the γ^K -contraction of $\mathcal{T}_K^\pi$ on the difference of its two images gives

$$\varepsilon_{\text{res}}^* \leq \varepsilon_{\text{anc}} + (1 + \gamma^K) \|U_* - B_{\text{anc}}\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}.$$

Since both $z^*(x, y)$ and $z_{\text{anc}}(x)$ lie in B_R ,

$$\begin{aligned} |U_*(x, y) - B_{\text{anc}}(x, y)| &= |V_\psi(z^*(x, y), x) - V_\psi(z_{\text{anc}}(x), x)| \\ &\leq L_V(R) \|z^*(x, y) - z_{\text{anc}}(x)\|_2 \leq L_V(R) D_{\text{anc}}(R). \end{aligned}$$

The special case $z_{\text{anc}}(x) \equiv 0$ uses $D_{\text{anc}}(R) \leq R$. $\square$

M.6 PROOF OF LEMMA L.2

Proof. For any (s, a) ,

$$\hat{Q}_V(s, a) - Q^\pi(s, a) = \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V(s') - V^\pi(s')].$$

If $s \in \mathcal{R}_\pi$ and $a \in \text{supp}(\pi(\cdot | s)) \cup \text{supp}(\pi_{\text{cand}}(\cdot | s))$, Definition I.2 gives $P(\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} | s, a) = 1$. Thus $|V(s') - V^\pi(s')| \leq \delta_V$ for transition-almost every s' , and hence $|\hat{Q}_V(s, a) - Q^\pi(s, a)| \leq \gamma \delta_V$.

Similarly,

$$|\mathbb{E}_{b \sim \pi(\cdot | s)} \hat{Q}_V(s, b) - \mathbb{E}_{b \sim \pi(\cdot | s)} Q^\pi(s, b)| \leq \gamma \delta_V,$$

since $b \sim \pi(\cdot | s)$ and the same almost-sure closure condition applies. Subtracting these two bounds yields $|\hat{A}_V(s, a) - A^\pi(s, a)| \leq 2\gamma \delta_V$. $\square$

M.7 PROOF OF THEOREM 6.5

Proof. The CPI bound (29) (derived in Appendix K) gives:

$$\eta(\pi_{\text{new}}) \geq L_\pi(\pi_{\text{new}}) - \frac{2\varepsilon_{\text{CPI}}\gamma}{(1-\gamma)^2}\alpha^2.$$

Now relate the true surrogate $L_\pi(\pi_{\text{new}})$ to the estimated surrogate $\hat{L}_\pi(\pi_{\text{new}})$:

$$L_\pi(\pi_{\text{new}}) - \hat{L}_\pi(\pi_{\text{new}}) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\pi, a \sim \pi_{\text{new}}(\cdot|s)} [A^\pi(s, a) - \hat{A}(s, a)].$$

For the mixture policy $\pi_{\text{new}} = (1-\alpha)\pi + \alpha\pi_{\text{cand}}$:

$$\begin{aligned} \mathbb{E}_{a \sim \pi_{\text{new}}(\cdot|s)} [A^\pi(s, a) - \hat{A}(s, a)] &= (1-\alpha) \mathbb{E}_{a \sim \pi(\cdot|s)} [A^\pi(s, a) - \hat{A}(s, a)] \\ &\quad + \alpha \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)} [A^\pi(s, a) - \hat{A}(s, a)]. \end{aligned}$$

By definition of advantage, $\mathbb{E}_{a \sim \pi(\cdot|s)} [A^\pi(s, a)] = 0$. By the centering assumption (18), $\mathbb{E}_{a \sim \pi(\cdot|s)} [\hat{A}(s, a)] = 0$. Hence the $(1-\alpha)$ term vanishes, and we obtain

$$\left| \mathbb{E}_{a \sim \pi_{\text{new}}(\cdot|s)} [A^\pi(s, a) - \hat{A}(s, a)] \right| = \alpha \left| \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)} [A^\pi(s, a) - \hat{A}(s, a)] \right| \leq \alpha \varepsilon_{A, \text{cand}}.$$

Why this is a tightening: Note that by Jensen's inequality and Assumption 6.4,

$$\varepsilon_{A, \text{cand}} = \sup_s \left| \mathbb{E}_{a \sim \pi_{\text{cand}}} [\hat{A}(s, a) - A^\pi(s, a)] \right| \leq \sup_s \mathbb{E}_{a \sim \pi_{\text{cand}}} [\hat{A}(s, a) - A^\pi(s, a)] \leq \varepsilon_A.$$

Thus the evaluation error contribution $\alpha \varepsilon_{A, \text{cand}} / (1-\gamma)$ is no larger than the naive bound $\varepsilon_A / (1-\gamma)$, and is smaller whenever $\alpha < 1$ or $\varepsilon_{A, \text{cand}} < \varepsilon_A$.

Taking expectation over $s \sim d_\pi$ yields

$$L_\pi(\pi_{\text{new}}) \geq \hat{L}_\pi(\pi_{\text{new}}) - \frac{\alpha}{1-\gamma} \varepsilon_{A, \text{cand}}.$$

Combining with the CPI bound gives (20). $\square$

M.8 PROOF OF COROLLARY B.1

Proof. The clock-complete state, frozen shared map h_n , and specified initial law define one augmented MDP for both action kernels. On its bounded measurable functions, $\bar{\mathcal{T}}_K^{\bar{\pi}}$ is a γ^K -contraction in the augmented closure norm. In this proof, $\|\cdot\|_\infty^{\text{adv}}$ abbreviates $\|\cdot\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}$. Thus

$$\begin{aligned} \|\tilde{U}_n - V^{\bar{\pi}}\|_\infty^{\text{adv}} &\leq \|\tilde{U}_n - \bar{\mathcal{T}}_K^{\bar{\pi}} \tilde{U}_n\|_\infty^{\text{adv}} + \|\bar{\mathcal{T}}_K^{\bar{\pi}} \tilde{U}_n - \bar{\mathcal{T}}_K^{\bar{\pi}} V^{\bar{\pi}}\|_\infty^{\text{adv}} \\ &\leq \bar{\varepsilon}_{\text{res}, n} + \gamma^K \|\tilde{U}_n - V^{\bar{\pi}}\|_\infty^{\text{adv}}. \end{aligned}$$

Rearrangement proves (37). Lemma L.2, applied on the augmented closure with exact centering conditional on $\bar{s}$, gives (38). The candidate-policy bias is no larger than this uniform advantage error. Under the exact pointwise mixture, substituting it into Theorem 6.5 proves (39). $\square$

For completeness, the optional slow-drift comparisons follow directly. Lemma B.7 and the Lipschitz head imply (44); adding and subtracting $\tilde{U}_*$ and applying the residual bound (1) proves (45). Finally, the reverse triangle inequality and Bellman contraction give

$$\begin{aligned} |\bar{\varepsilon}_{\text{res}, n} - \bar{\varepsilon}_{\text{res}, *}| &\leq \|(\tilde{U}_n - \tilde{U}_*) - (\bar{\mathcal{T}}_K^{\bar{\pi}} \tilde{U}_n - \bar{\mathcal{T}}_K^{\bar{\pi}} \tilde{U}_*)\|_\infty^{\text{adv}} \\ &\leq (1 + \gamma^K) \|\tilde{U}_n - \tilde{U}_*\|_\infty^{\text{adv}}, \end{aligned}$$

which proves (46).

Algorithm 2 UPI-TRM: single interaction + update step

```
1: Input:  $x, y_0, z_0, \gamma, n, K, \alpha, \tau, R$ 
2: Initialize  $\tilde{s}_0 \leftarrow (x, y_0, z_0)$ ;  $done \leftarrow \text{false}$ 
3: for  $k = 0$  to  $K - 1$  do
4:   if  $done$  then
5:      $r_k \leftarrow (\gamma - 1)C_{\max}$ ;  $\tilde{s}_{k+1} \leftarrow s_{\text{abs}}$  // Absorbing self-loop
6:   else
7:      $z_k^{(n)} \leftarrow (\Pi_R \circ f_\theta)^{\circ n}(z_k, y_k, x)$  // n inner steps
8:     Sample  $a_k \sim \pi_\phi(a \mid y_k, z_k^{(n)}, x)$ 
9:      $y_{k+1} \leftarrow \text{edit}(y_k, a_k; x)$ ; receive  $r_k$ 
10:     $z_{k+1} \leftarrow z_k^{(n)}$  // Carry latent (skip for episodic-z)
11:     $\tilde{s}_{k+1} \leftarrow (x, y_{k+1}, z_{k+1})$ 
12:    if terminated then
13:       $done \leftarrow \text{true}$ ;  $\tilde{s}_{k+1} \leftarrow s_{\text{abs}}$ 
14:    end if
15:  end if
16: end for
17: Compute target  $G^{(K)}$  via (24)
18: Update  $(\theta, \psi)$  by minimizing  $\mathcal{L}_{\text{val}}$ , then freeze the evaluator snapshot  $h_n$ 
19: Form advantages  $\hat{A}$  via centered estimator (see below)
20: Policy-head step with  $h_n$  frozen  $\rightarrow$  candidate  $\pi_{\text{cand}}$ 
21:  $\pi_{\text{new}} \leftarrow (1 - \alpha)\pi_\phi + \alpha\pi_{\text{cand}}$ 
22: Distill:  $\pi_\phi \leftarrow \arg \min_\phi \text{KL}(\pi_{\text{new}} \parallel \pi_\phi)$ 
23:  $\bar{\psi} \leftarrow \tau\bar{\psi} + (1 - \tau)\psi$  // Target network update
```

N IMPLEMENTATION DETAILS

N.1 ALGORITHM PSEUDOCODE

Algorithm 1 presents the full UPI-TRM update step in persistent- z form. For the episodic- z variant, skip line 10 (“Carry latent”) and reinitialize $z_k \leftarrow z_{\text{init}}(x, y_k)$ at the start of each loop iteration.

The frozen CPI comparison begins only after the value update and uses the same fixed recurrent map h_n for the current and candidate action kernels through the mixture step. Any subsequent recurrent-parameter update starts a new snapshot and is outside Theorem 6.5. The retained training repository is unavailable, so whether the headline implementation enforces this separation remains unaudited.

We maintain a replay buffer and sample mini-batches for value and policy updates. Latents in the buffer computed under older θ are stale, so we recompute $z^{(n)}$ for sampled states or use a slowly moving target encoder $f_{\bar{\theta}}$.

Compute and hardware. Each reported training or evaluation run uses a single device. The main training jobs and the trusted external PPO/A2C/DQN reruns, including the 5-step DQN variant, ran through GPU-enabled batch jobs; the evaluation scripts also support CPU fallback for analysis-only runs. Within the architecture-matched group, “matched” refers to the shared TRM backbone and checker-shaped reward, not to identical environment-interaction counts or constant-factor wall-clock cost. The nominal 20k outer-step protocol gives different interaction budgets across PPO, A2C, and DQN because each collects a different number of transitions per outer step; Appendix G.1 reports the missing-audit and artifact status, not validated equalized results. The baseline update rules also carry algorithm-specific overhead; for example, PPO and A2C evaluate policy and value heads in separate forwards through the TRM backbone during optimization, so wall-clock time per update can exceed UPI-TRM.

Advantage estimation. The theory uses the centered estimator (67); in practice $\text{GAE}/(\lambda)$ -return variants are also fine for variance reduction. The linear-in- α error term in Theorem 6.5 requires *exact* statewise centering $\mathbb{E}_{a \sim \pi}[\hat{A}(s, a)] = 0$, which holds deterministically only when the baseline is computed exactly as $B_V(s) = \mathbb{E}_{b \sim \pi(\cdot|s)}[\hat{Q}_V(s, b)]$ (e.g., by summation over a discrete action space). With a learned value baseline (as in standard GAE), centering is only approximate and the

deterministic bound degrades by the centering defect $\varepsilon_{\text{cent}}$; the naive $O(\varepsilon_A/(1 - \gamma))$ penalty is recovered only as a worst-case upper bound when $\varepsilon_{\text{cent}}$ and $\varepsilon_{A,\text{cand}}$ are both controlled by ε_A .

Operator-norm control and contraction details. We enforce contraction on the inner $z \rightarrow z$ recursion via explicit operator-norm clamping (periodic rescaling of linear weights using power-iteration estimates), followed by per-layer output scaling or an explicit scalar contraction factor to target $L_z < 1$. We estimate $\hat{L}_z$ on mini-batches by finite differences:

$$\hat{L}_z = \max_i \frac{\|f_\theta(z_i, y_i, x_i) - f_\theta(z'_i, y_i, x_i)\|}{\|z_i - z'_i\|}.$$

Enforcing forward-invariance. To ensure $z_{\text{init}}(x, y) \in \mathcal{Z}_{\text{inv}}$ and that iterates remain there:

- Use an initialization $z_{\text{init}}(x, y)$ with $\|z_{\text{init}}(x, y)\|_2 \leq R$, $\mathcal{Z}_{\text{inv}} = \{z : \|z\|_2 \leq R\}$. In the artifact implementation, the initial latent comes from learned latent seeds (optionally augmented by an initialization encoder) and, when $R > 0$, is projected into the same ball.
- After each update, apply Euclidean projection onto the ball: $z \leftarrow \Pi_R(z)$ with Π_R as in (23).
- Under the Euclidean norm this projection is 1-Lipschitz (non-expansive) and so preserves the contraction modulus.
- Equivalently, define $\tilde{f}_\theta := \Pi_R \circ f_\theta$ and unroll $\tilde{f}_\theta^n$: $z^{(t+1)} = \Pi_R(f_\theta(z^{(t)}, y, x))$ for $t = 0, \dots, n - 1$.

For persistent- z we monitor the fresh-initialization discrepancy $\hat{C}_{\text{fresh}}(n) = \max_t \|z_t^{(n)} - z_{\text{init}}(x, y_t)\|$. It measures separation from a fresh initialization; it is not a fixed-point tracking bound and is not a substitute for $C_{\text{drift}}(n)$ or the augmented residual.

Distillation. When distilling π_{new} to π_ϕ , we minimize:

$$\mathcal{L}_{\text{distill}}(\phi) = \mathbb{E}_{\tilde{s} \sim \text{buffer}} [\text{KL}(\pi_{\text{new}}(\cdot | \tilde{s}) \| \pi_\phi(\cdot | \tilde{s}))].$$

This is a heuristic outside the theory. A finite-batch $\text{KL}(\pi_{\text{new}} \| \pi_\phi)$ should be reported when both policies are retained, but it would still not certify exact deployment. That diagnostic is not available for the headline run, so its distillation gap is unmeasured rather than zero.

O MAPPING TO ORIGINAL TRM ARCHITECTURE

We provide an explicit mapping between our formalism and the original TRM implementation (Jolicoeur-Martineau, 2025).

Our Formalism	TRM Implementation	Notes
$z \in \mathcal{Z}$	Hidden state	Latent reasoning state
$y \in \mathcal{Y}$	Output grid/sequence	Candidate solution
f_θ	Recurrent update	Single update step
n (inner steps)	Recursion depth	Fixed or adaptively chosen evaluation budget
$z_{\text{init}}(x, y)$	Initial hidden	Learned (projected when $R > 0$)
$\pi_\phi(a y, z, x)$	Task head (reinterpreted)	TRM’s task head produces the refined answer; we reframe the answer update as a discrete edit policy
$\text{edit}(y, a; x)$	Continuous refinement	TRM: continuous y -update via another forward pass; we discretize into edit actions
Episodic- z	Not in original	Our simplification
Persistent- z	Original TRM	z carried across edits
Supervised objective	Training signal	Replaced by RL reward
Adaptive halting	Early stopping	Can be related to variable n

The key differences:

1. **Training signal:** The original TRM is trained with a supervised loss; we replace this with checker-based rewards and a CPI-style policy-improvement loop.

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- 2160 2. **Episodic- z vs. persistent- z :** We analyze both; the original TRM is persistent- z .
2161
2162 3. **Halting:** The original TRM uses an adaptive halting mechanism; our analysis fixes n for
2163 tractability and treats n as an analysis parameter rather than a learned quantity.

2164 P ETHICS AND BROADER IMPACT

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2167 **Ethics statement.** The paper studies checker-trained Tiny Recursive Models on synthetic Sudoku
2168 tasks. We use no personal data, no human subjects, no crowdsourcing, no deployed decision system.
2169 The main ethical risk is rhetorical: narrow verifier-feedback results on puzzle tasks can be overstated
2170 as evidence about general reasoning or high-stakes automation. We scope the empirical claim to
2171 controlled Sudoku, do not include the 9×9 feasibility sweep as a result, and state in the limitations
2172 paragraph that the bound does not certify improvement.

2173 **Broader impact.** The methodological contribution is to separate internal evaluator depth from
2174 outer policy improvement, which may help analyses of verifier-feedback systems with bounded
2175 internal compute. The main risk is misinterpretation of puzzle-domain results as broader claims about
2176 reasoning capability; we therefore avoid deployment recommendations and keep the paper’s claims
2177 narrow.

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